TRACE X: Cross-System Fraud Intelligence and Investigation Platform

Azentio AI Hackathon, Challenge #01 · Team Byte Me

Item	Detail
Live application	https://trace-x-rouge-rho.vercel.app
Source code	https://github.com/AzentioHackathon-2026/Digital-Fraud-Investigation-System-byte-me
Sign-in	`test@agentio.com` (analyst) · `thanushgarimella@gmail.com` (administrator / supervisor) · password `agentio@123`
Report date	31 October 2026

1. Executive summary

TRACE X ingests records from core banking, lending, insurance and digital channels and detects fraud that only becomes visible when those systems are read together. It explains every score with the records behind it and gives investigators a complete workspace. On top of that sits an agent team that investigates cases, learns from analysts' corrections and earns limited autonomy only where its measured accuracy justifies it.

Headline results. All numbers are reproducible with the commands in section 11. Every dataset is synthetic with planted fraud, and seeds 201 and 202 were kept sealed until final evaluation.

Measure	Result
Planted fraud detected (8 detectors)	**18 / 18 on every seed**, including sealed seeds 201 and 202
Legitimate lookalikes kept Low	**10 / 10 on every seed**
Precision of actionable alerts	87 – 100 % across seeds (sealed: 90.9 % and 91.7 %)
Graph neural network, sealed test	ROC-AUC **0.997**, average precision **0.879**; at the validated threshold, precision **0.94** and recall **1.00**
Agent team accuracy, sealed seeds	**18 / 19 cases** (94.7 %), **99 %** of cited evidence verified in the database
Automated tests	**111** unit and integration tests, **20** Level 1 and **32** Level 2 end-to-end checks; all pass
Detection speed	about 160 ms for 10,000 records (6 source tables)

Requirements checklist

Live application	https://trace-x-rouge-rho.vercel.app
Source code	https://github.com/AzentioHackathon-2026/Digital-Fraud-Investigation-System-byte-me
Analyst login	`test@agentio.com` · password `agentio@123`
Supervisor / admin login	`thanushgarimella@gmail.com` · password `agentio@123`

Level 1: Detect, score and manage fraud

- [x] CSV ingestion (parser, header aliases, field normalisation)
- [x] Excel ingestion (multi-sheet XLSX)
- [x] JSON ingestion (array and object-of-arrays)
- [x] Mandatory-field validation (row-level errors, warnings, rejections)
- [x] Duplicate detection and import summaries
- [x] Data lineage (batch ID, source file, row, hash)
- [x] FD-01 Suspicious transfers (amount, velocity, new beneficiary, baselines)
- [x] FD-02 Shared-device fraud (coordinated identities)
- [x] FD-03 Account takeover (failed logins, new device, sensitive activity)
- [x] Legitimate lookalikes kept Low (household and shared-device context)
- [x] Additional scenarios: loan, identity and insurance fraud
- [x] Multi-indicator risk scoring (weights, mitigations, floors)
- [x] Explainable risk linked to evidence
- [x] Fraud dashboard (counts, severity, categories, latest findings)
- [x] Alert search and filters (customer, transaction, product, risk, date, detector)
- [x] Source evidence and event timelines
- [x] Case queue (ID, customer, risk, alerts, status, evidence)
- [x] Requires Investigation decision
- [x] False Positive decision with reason
- [x] Original alert preserved (immutable)
- [x] Repeatable detection (run IDs, idempotent re-runs)

Level 2: Investigate, correlate and manage cases

1. Cross-system correlation

- [x] Customers, accounts and transactions linked
- [x] Devices, IP addresses and phone numbers linked
- [x] Loan applications and payout accounts linked
- [x] Insurance policies and claims linked
- [x] Banking, lending and insurance correlated
- [x] Links only through matching identifiers
- [x] Related findings consolidated into one investigation
- [x] Relationships behind each correlation explained

2. Relationship analysis

- [x] Interactive relationship network (Cytoscape)
- [x] Customer, account, device, application, transaction and claim nodes
- [x] Evidence-backed connections
- [x] Select an entity to inspect its records
- [x] Select a relationship to inspect its evidence
- [x] Graph ↔ source records ↔ timeline navigation
- [x] Fraud paths and connected entities highlighted

3. Case consolidation

- [x] Related alerts grouped into one case
- [x] Reason for grouping explained
- [x] Shared identifiers and linking evidence shown
- [x] Original alerts preserved
- [x] Analyst-driven case splitting
- [x] Reason required to split
- [x] Split and supersession history kept
- [x] Split persists across detection re-runs (tested)

4. Investigation workspace

- [x] Detailed case timeline
- [x] Affected customers and entities
- [x] Fraud indicators and explanations
- [x] Source evidence and references
- [x] Risk and confidence
- [x] Investigation notes
- [x] Requires Investigation
- [x] False Positive with reason
- [x] Request Additional Information
- [x] Escalate to Specialist / Supervisor
- [x] Recommend Temporary Hold
- [x] Recommend Enhanced Verification
- [x] Record Investigation Conclusion
- [x] Investigator identity and timestamps retained
- [x] Human approval for customer-impacting actions

5. Evidence and audit trail

- [x] Original source records preserved
- [x] Fraud indicators stored
- [x] AI assessments stored
- [x] Analyst notes and corrections stored
- [x] Previous risk classification stored
- [x] Decisions with time and user identity
- [x] Append-only investigation history
- [x] Superseded cases preserved
- [x] Structured JSON case report
- [x] Downloadable PDF case report
- [x] Supporting evidence included in reports

6. Prioritisation

- [x] Prioritised by risk and severity
- [x] Estimated financial exposure
- [x] Linked alerts and cross-system relationships considered
- [x] Evidence confidence displayed
- [x] Monitored vs actionable findings
- [x] Prioritised case queue
- [x] Exposure at case and dashboard level
- [x] Low-evidence cases routed to Needs More Evidence (tested)

Built beyond Level 1 and Level 2

- [x] FD-08 Dormant account reactivation
- [x] Isolation Forest anomaly model
- [x] Graph neural network on the customer graph
- [x] FD-GNN monitored model-only findings
- [x] Four-agent team: orchestrator, hypothesis, SQL, reviewer
- [x] LLM investigation (gpt-4o-mini)
- [x] Read-only, guarded SQL tools
- [x] Evidence-ID verification (fabricated IDs dropped)
- [x] Deterministic agent fallback
- [x] Lessons learned from analyst feedback
- [x] Per-category agent trust scores
- [x] Trust-gated autonomy L0–L3 (threshold crossing tested)
- [x] Kill switch and autonomous allow-list
- [x] Human-in-the-loop approvals
- [x] Agent Trust dashboard
- [x] Agent evaluation (deterministic vs LLM)
- [x] Held-out ML benchmark (sealed seeds 201, 202)
- [x] Feedback-based policy suggestions (never auto-applied)
- [x] Session telemetry and payee-name check fields
- [x] Workspace activity audit
- [x] Persistent Turso database
- [x] Production reset safeguards
- [x] Responsive mobile screens and dark mode
- [x] Model card and agent evaluation docs

21 Level 1 items, 57 Level 2 items and 24 additional features, all implemented and verified by the automated tests and end-to-end checks in section 11 or the screenshots in this report.

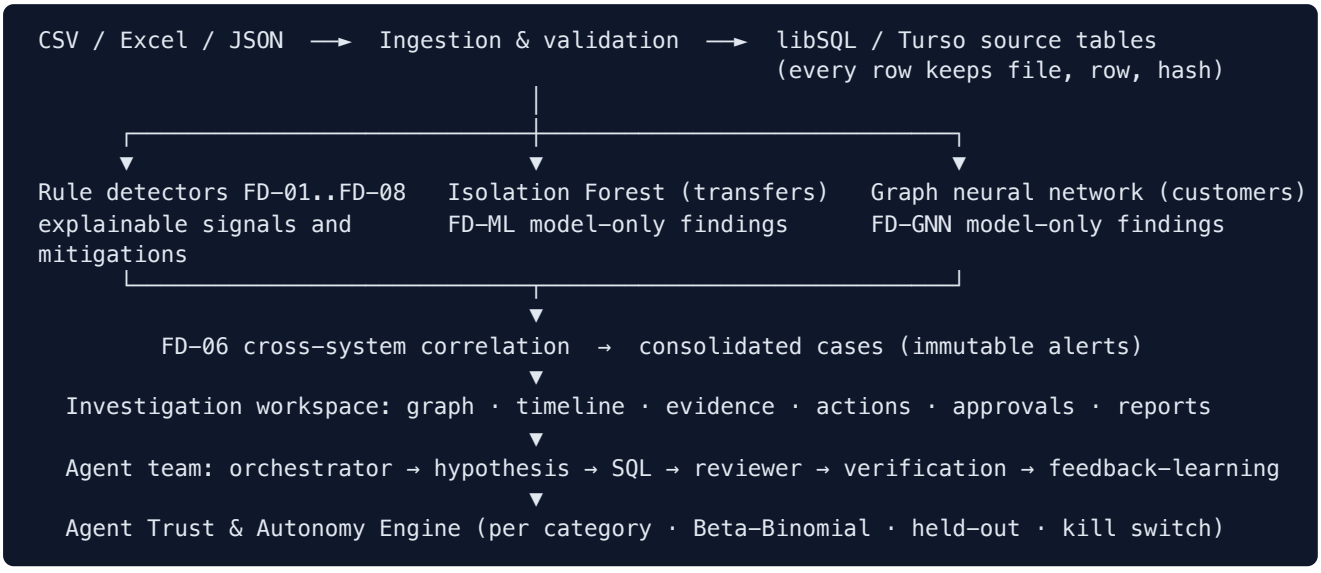
2. The problem and our approach

Fraud rarely stays inside one system. A loan ring opens accounts in banking, applies in lending and cashes out through insurance. An account takeover starts in the digital channel and ends in a transfer to a mule. Each step looks minor where it happens, so product-siloed controls miss the pattern.

TRACE X answers four questions for every case:

1. **What happened?** A chronological journey across systems, with each event's role marked.
2. **Why is it suspicious?** Weighted signals and legitimate-context mitigations, each citing source records.
3. **How sure are we, and how much is at stake?** Risk, evidence confidence, financial exposure, priority and agent trust, kept as separate numbers.
4. **What should happen next, and who decides?** Recommended actions, an approval workflow for anything customer-impacting, and an immutable audit trail.

3. Architecture



Stack: Next.js 16 (App Router), TypeScript, libSQL/Turso, Cytoscape.js, zod, Tailwind CSS, and the OpenAI API (gpt-4o-mini) for the agent team. Deployed on Vercel. All models (Isolation Forest, GNN, Beta-Binomial trust) are implemented in the repository with no external ML service.

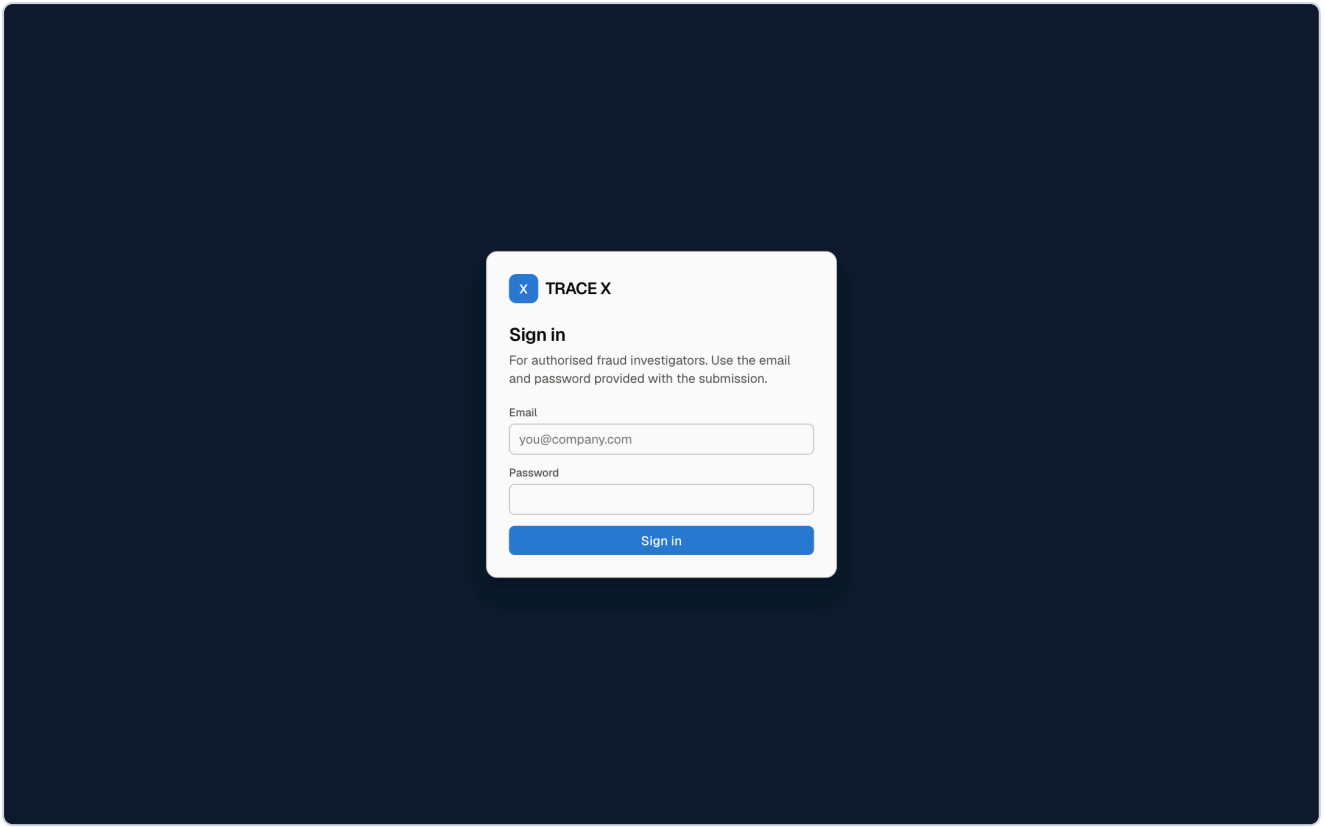
Integrity by design:

- Alerts, cases, decisions, actions, splits, agent feedback and the autonomous-action log are append-only, enforced by database triggers rather than application code.
- Alert identities are content hashes, so re-running detection is idempotent.
- Reset is the only path that removes data, and the activity log survives it.

4. Level 1 capabilities

4.1 Sign-in and roles

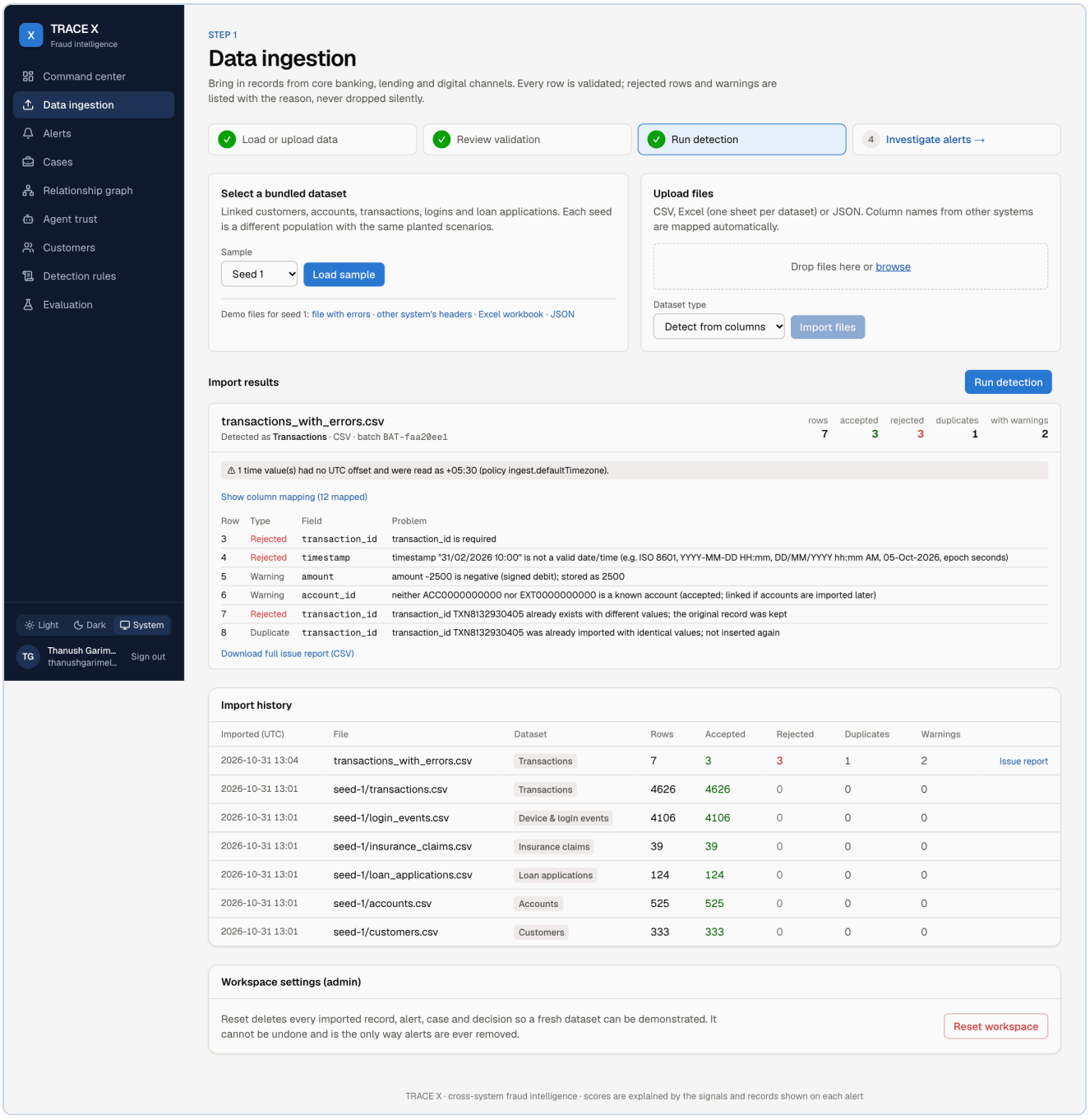
Two roles exist. An analyst investigates and records actions. An administrator (supervisor) can also approve customer-impacting recommendations, reset the workspace and control agent autonomy. Sessions are HMAC-signed cookies; passwords are stored as scrypt hashes.



Sign-in

4.2 Data ingestion and validation

- **Formats and headers:** CSV, Excel and JSON for all six record types: customers, accounts, transactions, login events, loan applications and insurance claims. Headers are matched through aliases, so another system's column names work ("Cust ID", "Login Time", "Auth Result").
- **Dates and amounts:** date order is detected per column; currency symbols, Dr/Cr notation and negative amounts are normalised.
- **Row-level outcomes:** every row is accepted, accepted with a warning, rejected or flagged as a duplicate, with the reason shown. The screenshot shows a deliberately broken file: one valid row, plus a missing ID, an impossible date, a negative amount, unknown accounts and duplicates.



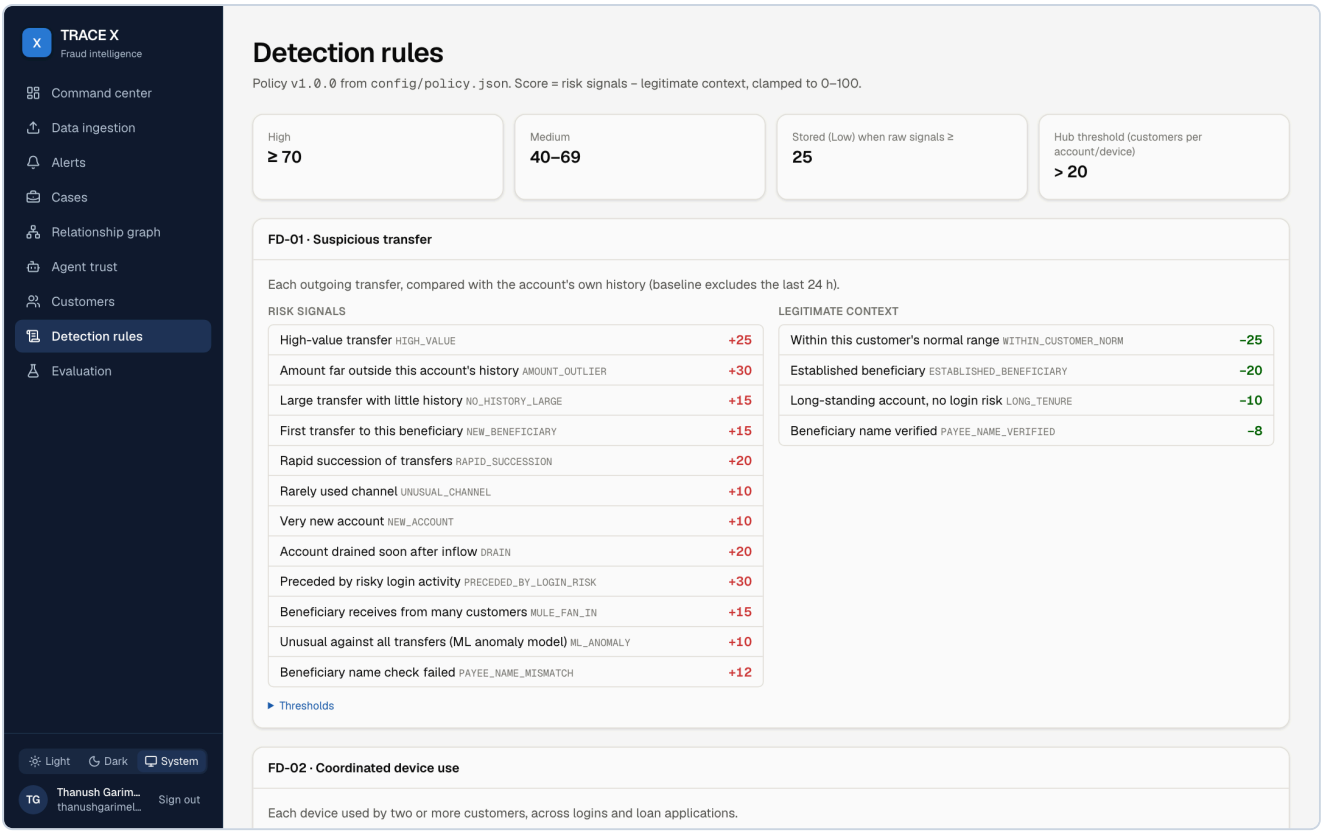
Ingestion validation

4.3 Detection engine

Eight rule detectors produce explainable findings. Each finding carries weighted **signals** (adding risk) and **mitigations** (legitimate context removing risk), and every signal cites the records it used.

Detector	Detects	Example signals	Example mitigations
FD-01 Suspicious transfer	Unusual amounts, new beneficiaries, drained new accounts, mule fan-in	Amount outlier (robust z), new beneficiary, drain after inflow, payee-name mismatch	Within customer's norm, established beneficiary, payee name verified
FD-02 Coordinated device use	Many new, unrelated profiles on one device	Three or more profiles, short first-use window, shared payout, all recently onboarded	Same household, long-standing shared use
FD-03 Account takeover	Failed-login burst or phished credentials, then access and money movement	Failed burst, new device, new location, SIM swap, remote-access tool, behaviour mismatch	Known device, behaviour consistent
FD-04 Identity / synthetic	Shared national ID, phone or email; implausible profiles	ID reuse under another name, shared contact, clustered onboarding, implausible attributes	Same household, no ID reuse
FD-05 Loan application fraud	Application rings	Shared payout account, shared device, velocity, income mismatch	Same household, no shared payout
FD-06 Cross-system	Alerts from several detectors linked by shared entities	Strongest member, independent detectors, multiple products, bridging entities	—
FD-07 Insurance claim fraud	Duplicate invoices and incidents, early claims, payouts to others	Duplicate document or incident, early claim, near sum insured, shared payout, late reporting	Household payout, long-standing policy
FD-08 Dormant reactivation	Long-dormant accounts suddenly moving money	Dormant then active, large inflow, pass-through, new payees	Only payees paid before dormancy, small amounts
FD-ML Isolation Forest	Transfers that are statistical outliers but match no rule	14 behavioural features, seeded	—
FD-GNN Graph neural network	Customers whose records and neighbours look like known fraud	26 features × 3 hops	—

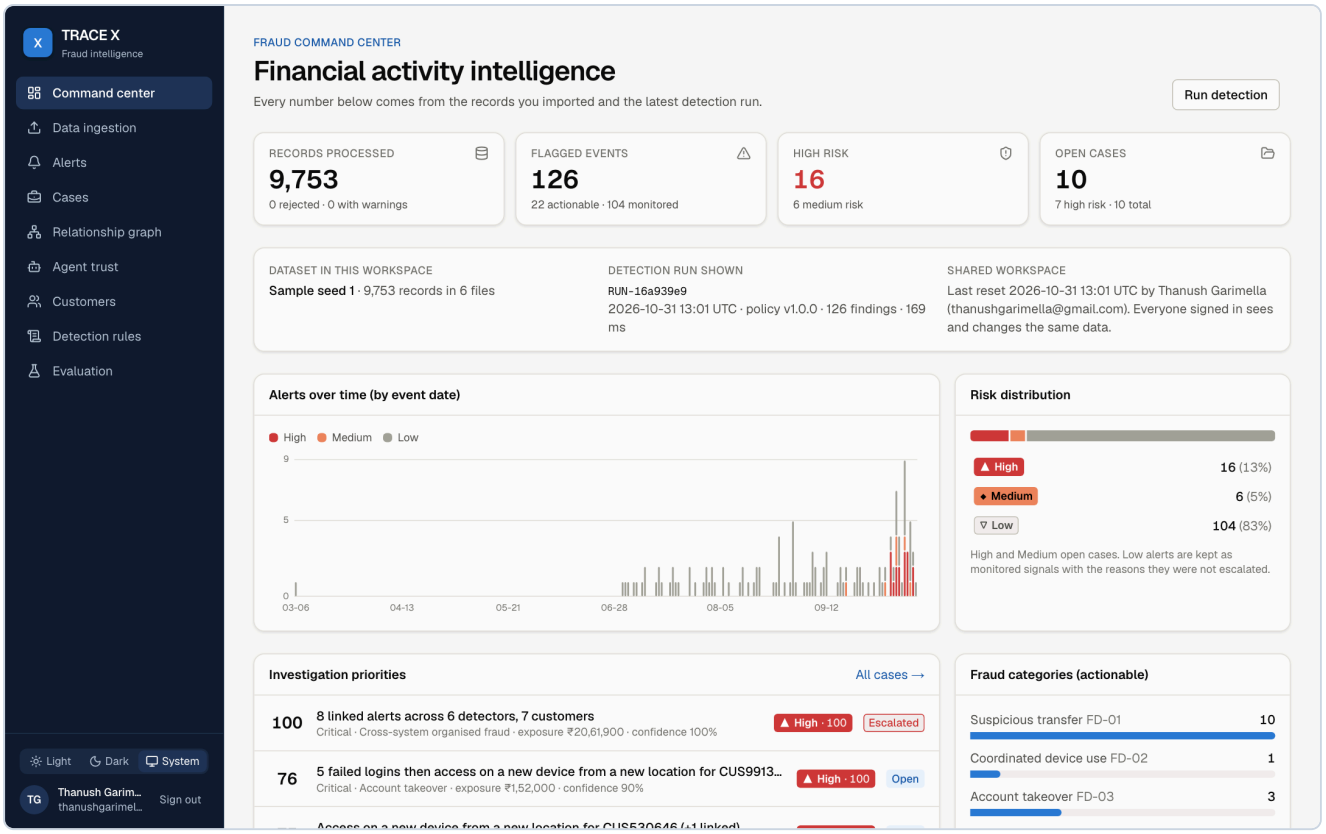
Scoring: score = Σ signals − Σ mitigations, clamped to 0–100, with severity floors for decisive patterns. Levels are High ≥ 70, Medium 40–69 and Low < 40. High and Medium alerts open cases. Low alerts are kept as "monitored", together with the reason they were not escalated.



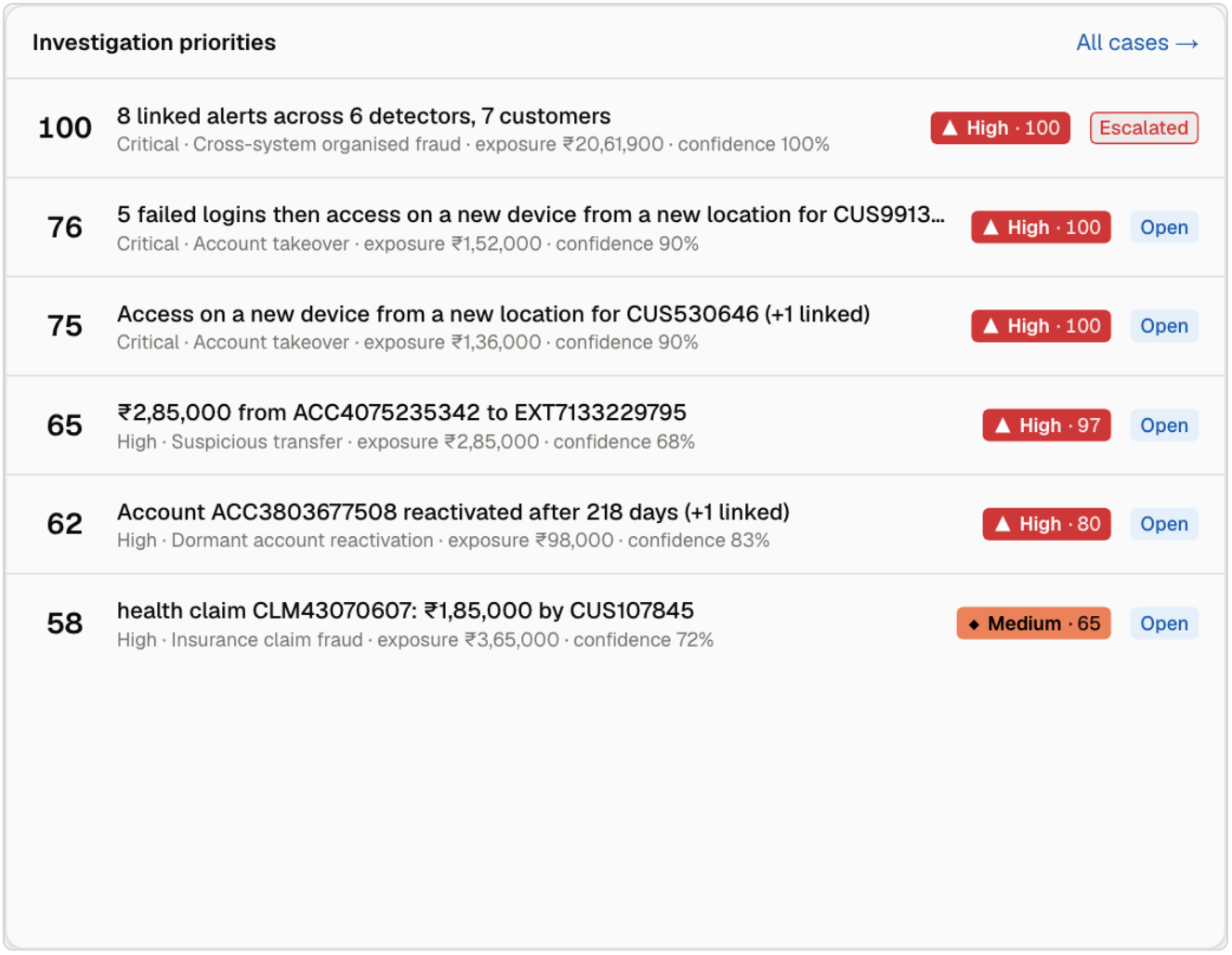
Detection rules page

4.4 Command center (dashboard)

The dashboard shows everything computed from the workspace: records processed, flagged events, risk distribution and alerts over time. It also shows a priority-ordered investigation list (with exposure and confidence), the agent trust overview, the dataset and detection run displayed, and a workspace activity log recording who loaded, reset, decided or approved what.


TRACE X
Fraud intelligence
Command center
Data ingestion
Alerts
Cases
Relationship graph
Agent trust
Customers
Detection rules
Evaluation
Light
Dark
System
Thanush Garim...
thanushgarimel...
Sign out
FRAUD COMMAND CENTER
Financial activity intelligence
Every number below comes from the records you imported and the latest detection run.
Run detection
RECORDS PROCESSED
9,753
0 rejected · 0 with warnings
FLAGGED EVENTS
126
22 actionable · 104 monitored
HIGH RISK
16
6 medium risk
OPEN CASES
10
7 high risk · 10 total
DATASET IN THIS WORKSPACE
Sample seed 1 · 9,753 records in 6 files
DETECTION RUN SHOWN
RUN-16a939e9
2026-10-31 13:01 UTC · policy v1.0.0 · 126 findings · 169 ms
SHARED WORKSPACE
Last reset 2026-10-31 13:01 UTC by Thanush Garimella (thanushgarimella@gmail.com). Everyone signed in sees and changes the same data.
Alerts over time (by event date)
High
Medium
Low
9
5
0
03-06
04-13
05-21
06-28
08-05
09-12
Risk distribution
High
16 (13%)
Medium
6 (5%)
Low
104 (83%)
High and Medium open cases. Low alerts are kept as monitored signals with the reasons they were not escalated.
Investigation priorities
All cases →
100
8 linked alerts across 6 detectors, 7 customers
Critical · Cross-system organised fraud · exposure ₹20,61,900 · confidence 100%
High · 100
Escalated
76
5 failed logins then access on a new device from a new location for CUS9913...
Critical · Account takeover · exposure ₹1,52,000 · confidence 90%
High · 100
Open
Fraud categories (actionable)
Suspicious transfer FD-01
10
Coordinated device use FD-02
1
Account takeover FD-03
3

Dashboard


Investigation priorities
All cases →
100
8 linked alerts across 6 detectors, 7 customers
Critical · Cross-system organised fraud · exposure ₹20,61,900 · confidence 100%
High · 100
Escalated
76
5 failed logins then access on a new device from a new location for CUS9913...
Critical · Account takeover · exposure ₹1,52,000 · confidence 90%
High · 100
Open
75
Access on a new device from a new location for CUS530646 (+1 linked)
Critical · Account takeover · exposure ₹1,36,000 · confidence 90%
High · 100
Open
65
₹2,85,000 from ACC4075235342 to EXT7133229795
High · Suspicious transfer · exposure ₹2,85,000 · confidence 68%
High · 97
Open
62
Account ACC3803677508 reactivated after 218 days (+1 linked)
High · Dormant account reactivation · exposure ₹98,000 · confidence 83%
High · 80
Open
58
health claim CLM43070607: ₹1,85,000 by CUS107845
High · Insurance claim fraud · exposure ₹3,65,000 · confidence 72%
Medium · 65
Open

Investigation priorities

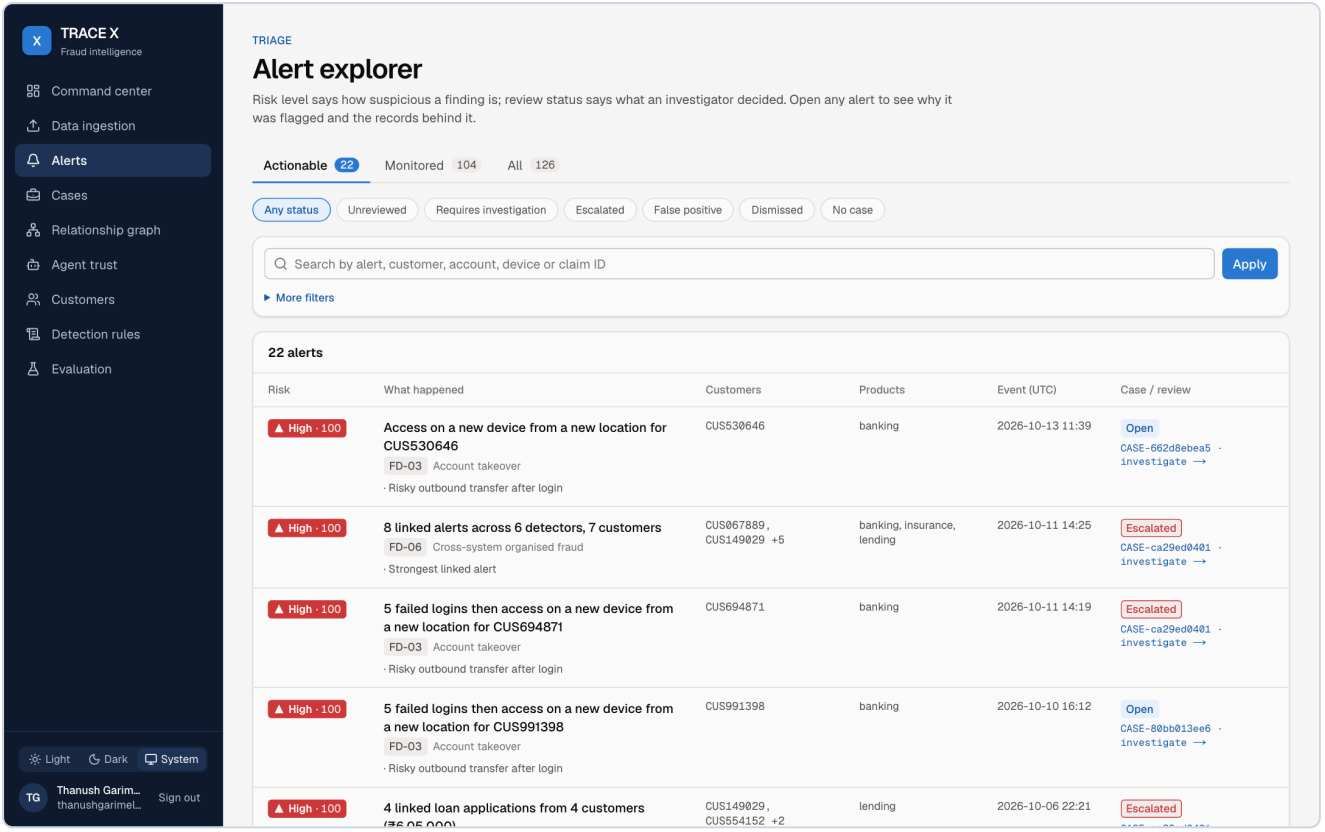
Recent workspace activity

2026-10-31 13:02 UTC	FEEDBACK	rejected agent recommendation on CASE-6910bbf4d1: trust 0.062 → 0.058	Thanush Garimella (thanushgarimella@gmail.com)
2026-10-31 13:02 UTC	AGENT	llm-multi-agent assessed CASE-6910bbf4d1: Suspicious transfers	Thanush Garimella (thanushgarimella@gmail.com)
2026-10-31 13:02 UTC	APPROVAL	approved recommendation #3 on CASE-ca29ed0401	Thanush Garimella (thanushgarimella@gmail.com)
2026-10-31 13:02 UTC	ACTION	add note on CASE-ca29ed0401	Test Investigator (test@agentio.com)
2026-10-31 13:02 UTC	ACTION	recommend hold on CASE-ca29ed0401	Test Investigator (test@agentio.com)
2026-10-31 13:02 UTC	ACTION	escalate supervisor on CASE-ca29ed0401	Test Investigator (test@agentio.com)
2026-10-31 13:02 UTC	ACTION	request information on CASE-ca29ed0401	Test Investigator (test@agentio.com)
2026-10-31 13:02 UTC	FEEDBACK	accepted agent recommendation on CASE-ca29ed0401: trust 0.009 → 0.034	Test Investigator (test@agentio.com)

Workspace activity log

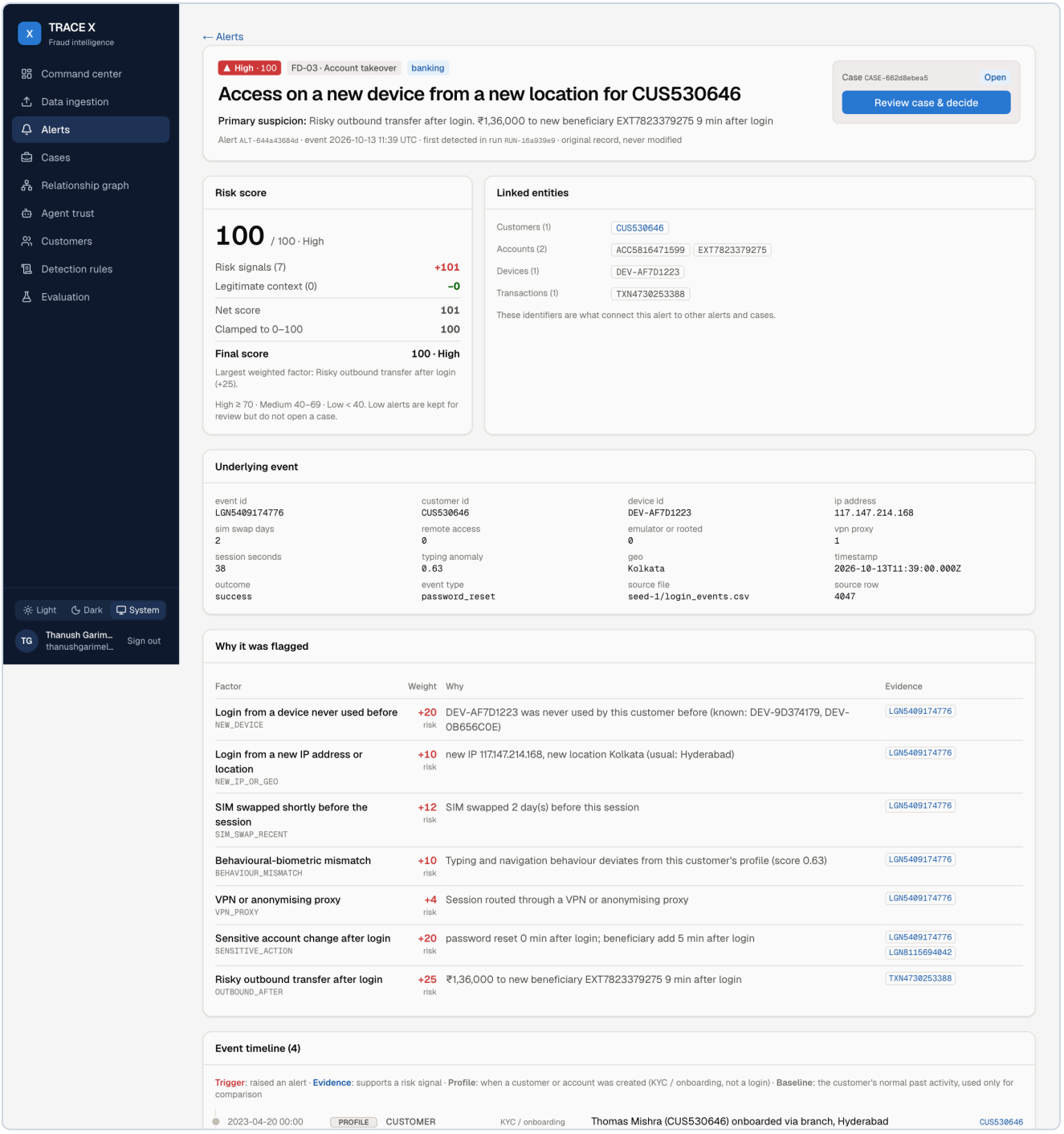
4.5 Alert explorer

Tabs split actionable alerts from monitored ones. Filters cover customer, transaction, product, risk level, detector, review status, date range and free-text search. Each row shows the strongest reason, any legitimate context, the case it belongs to, and a link to investigate.



Alerts

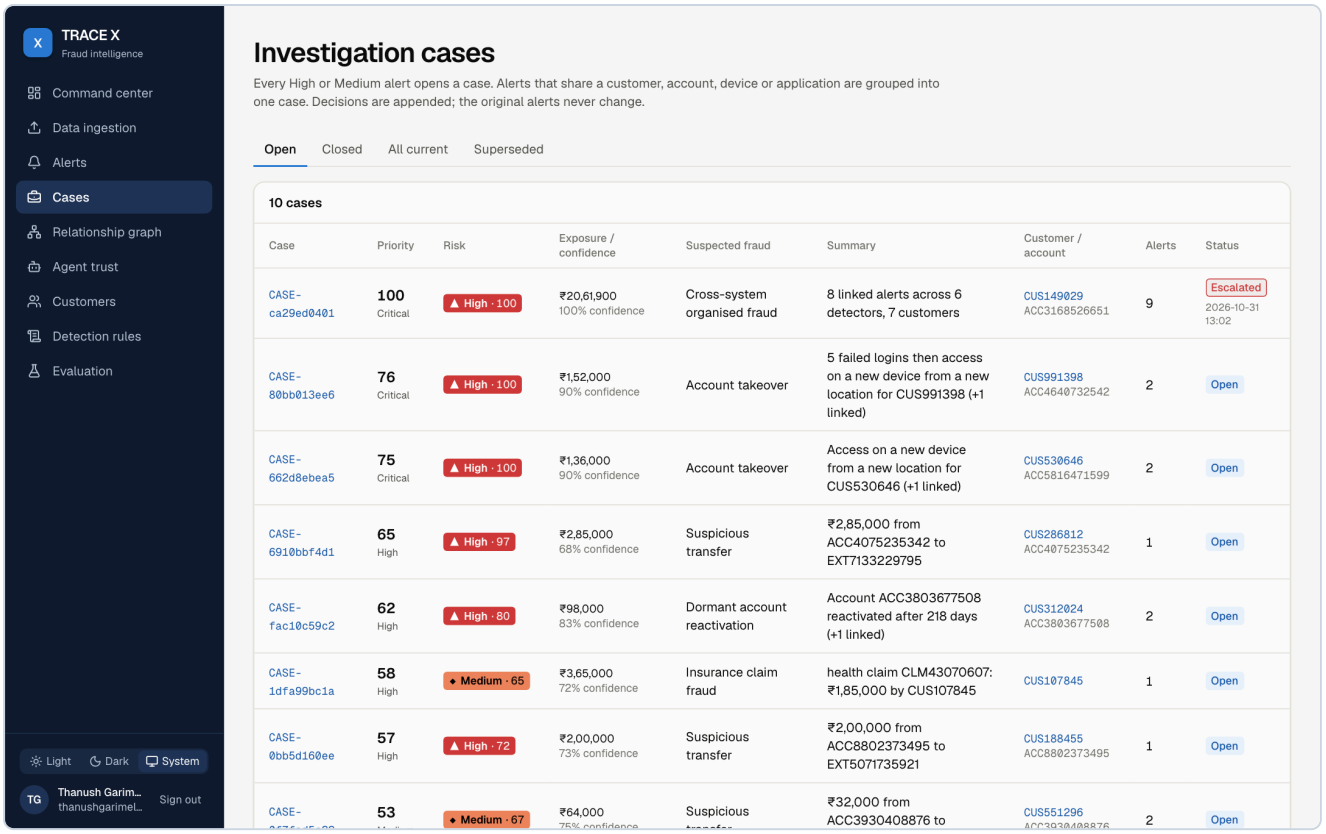
The alert detail page shows the scoring breakdown, linked entities, the timeline and every source record. Below is an account-takeover alert that uses the new session-telemetry signals (SIM swap, remote-access tool, behaviour mismatch).



Alert detail

4.6 Case queue

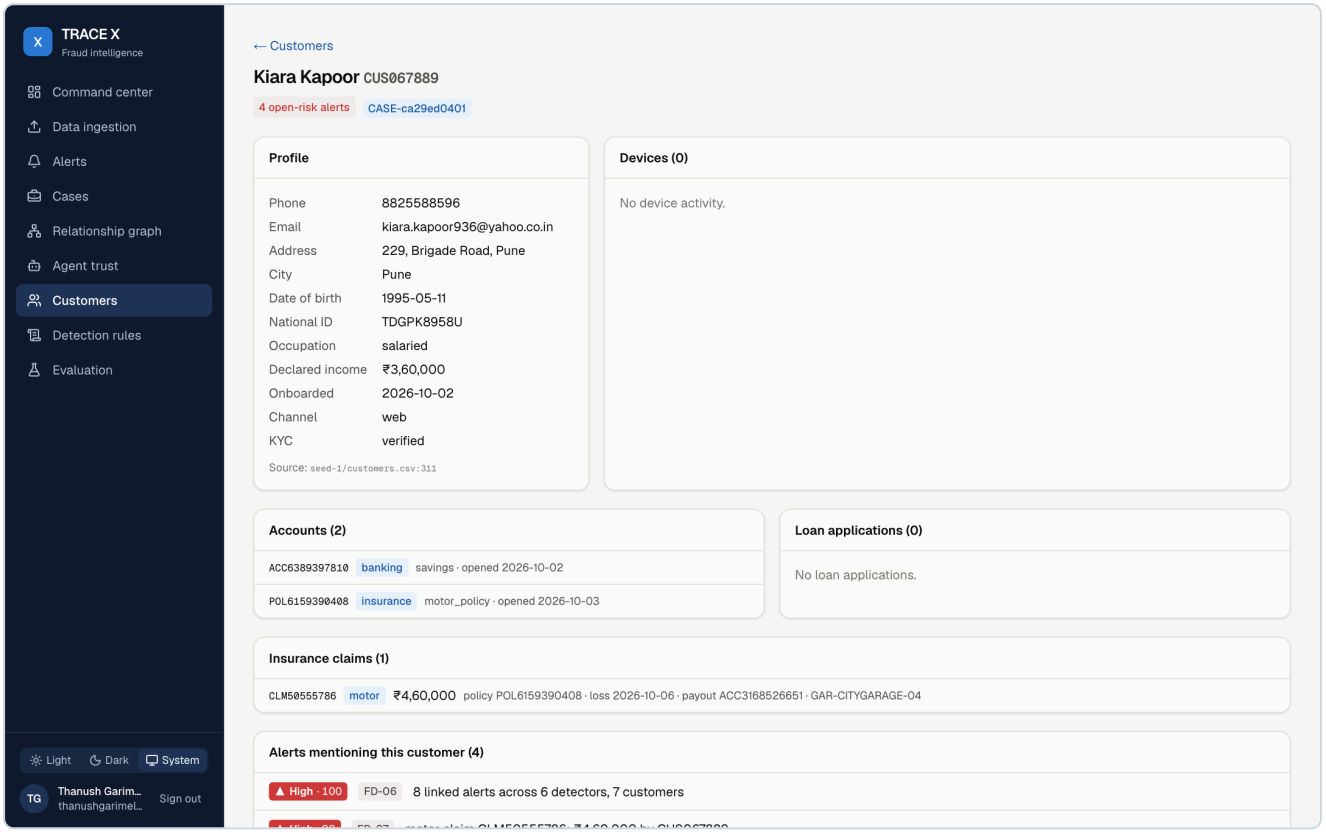
Cases are ordered by investigation priority, not by raw score. Each row shows priority tier, risk, exposure, evidence confidence, category, customer, alert count and review status. Cases that a newer detection run superseded keep their decisions under a separate tab.



Cases

4.7 Customer profile

A customer's profile brings together their accounts, devices, logins, transfers, applications, claims, alerts and cases.



Customer profile

5. Level 2 capabilities

5.1 Cross-system fraud correlation (requirement 1)

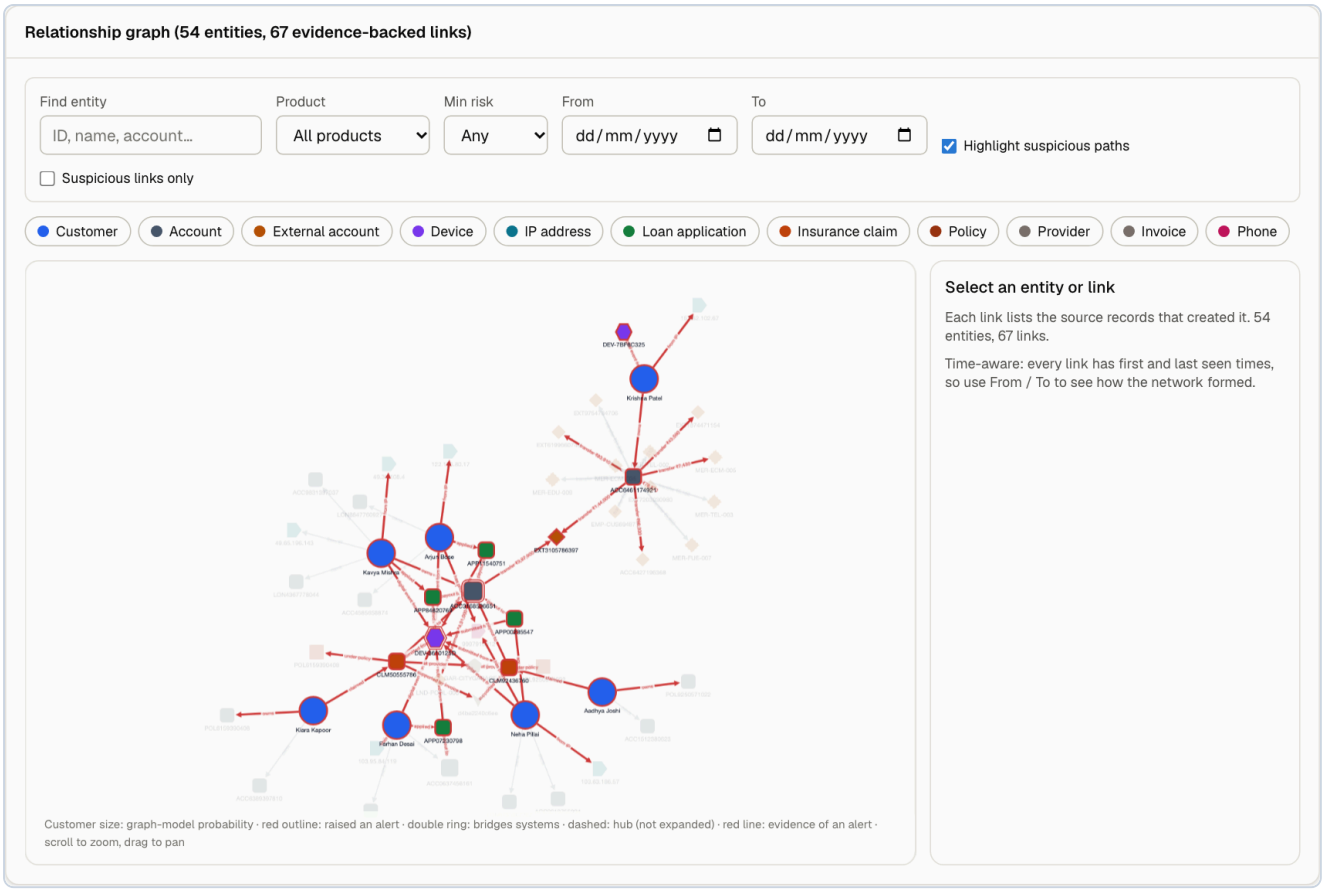
FD-06 links alerts through **every required entity type**: customer, account or beneficiary, device, IP address, phone, loan application, transaction, insurance policy and claim. Invoice documents link alerts as well.

- **Grounded links:** every link comes from a matching identifier in the source records.
- **Time-aware:** phone, IP, policy and document links join alerts only within a 45-day window.
- **Hub attenuation:** high-degree hubs such as merchant accounts, kiosk devices and shared IPs are excluded, so unrelated customers are not merged.
- **Tested:** each entity type has its own test.

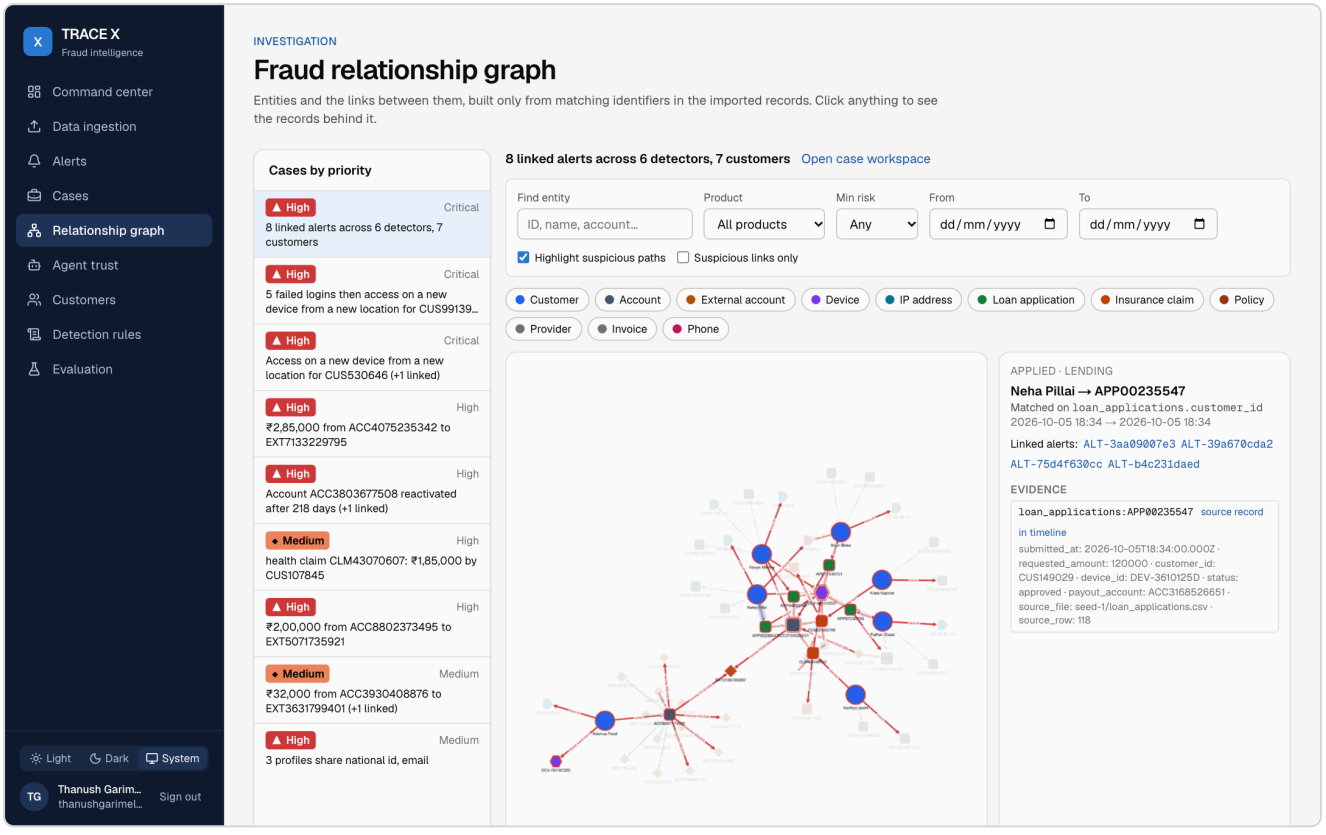
5.2 Interactive relationship graph (requirement 2)

Built with Cytoscape.js:

- **Entities:** 13 types, each with its own shape and colour, joined by directed, labelled edges such as owns, transfer (with amount), applied, payout to, claimed, under policy, invoice, login from and has phone.
- **Highlighting:** suspicious paths are drawn in red. Bridge entities that connect several systems get a double ring, and hubs are drawn dashed and not expanded. A customer's node size is their graph-model probability.
- **Inspection:** clicking a node or edge shows the exact matching field (for example `loan_applications.payout_account`), first and last seen times, the amount, linked alerts and the source records, with links to the record table and to the event in the timeline.
- **Filters:** entity type, product, minimum risk, date range, entity search and "suspicious links only".
- **No invented edges:** a test resolves every edge's evidence against the database.



Case graph



Graph page with a link selected

5.3 Intelligent case consolidation and splitting (requirement 3)

The case page explains **why alerts were grouped**: each shared entity, the source field that matched, which alerts it links, how far apart in time they were and whether that falls inside the correlation window. It also lists the hubs that were excluded.

Analysts can **split** wrongly grouped alerts into a new case, giving a reason:

- **Nothing is lost:** original alerts, evidence and decisions are untouched, and the move is an append-only record.
- **The split sticks:** a later detection run keeps the two groups apart, and the analyst's case IDs stay stable.
- **The old summary goes:** the combined cross-system summary alert is retired, because it no longer describes either half.

Why these alerts were consolidated

8 alerts from 6 detector(s) were grouped because they share 22 entities found in the source records.

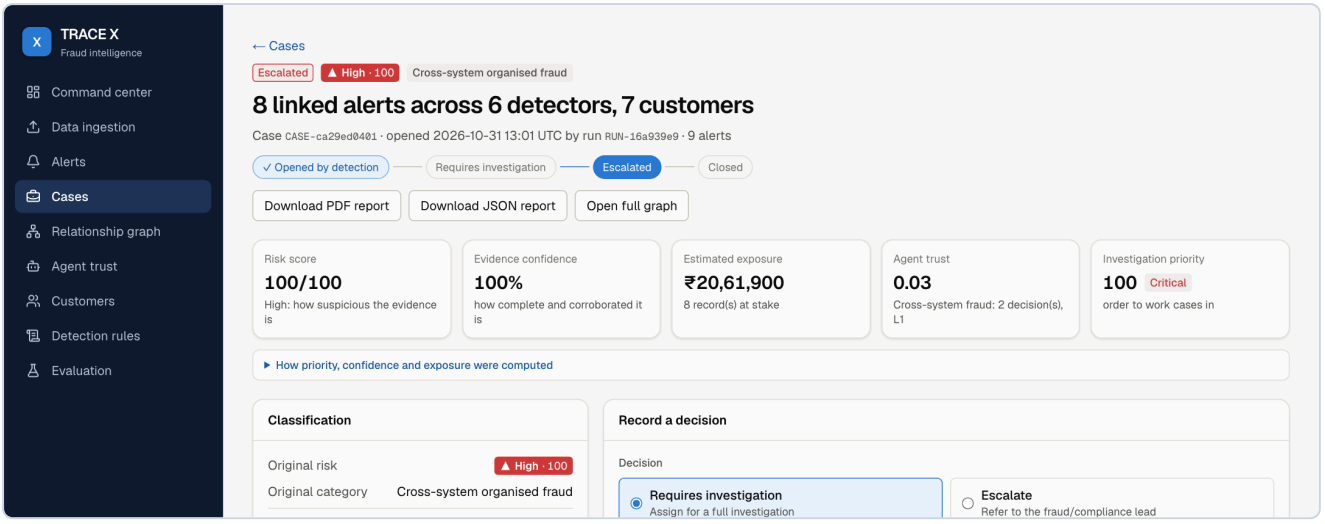
Shared entity	Matching source field	Linked alerts	Time relationship
Account ACC3168526651	account_id (owner, beneficiary or payout account)	FD-05 FD-01 FD-07 FD-02 FD-07	1.5 day(s) apart
Phone 9997915012	customers.phone_norm	FD-05 FD-07 FD-02 FD-07 FD-04	5.4 day(s) apart within 45-day window
Device DEV-3610125D	device_id	FD-05 FD-07 FD-02 FD-07	1.5 day(s) apart
Account EXT3105786397	account_id (owner, beneficiary or payout account)	FD-03 FD-01 FD-01	3.7 day(s) apart
Claim CLM50555786	insurance_claims.claim_id	FD-07 FD-02 FD-07	1 day(s) apart
Claim CLM92436760	insurance_claims.claim_id	FD-07 FD-02 FD-07	1 day(s) apart
Policy POL6159390408	insurance_claims.policy_id	FD-07 FD-02 FD-07	1 day(s) apart within 45-day window
Invoice document d4be2240c6eee7f4	insurance_claims.document_hash	FD-07 FD-02 FD-07	1 day(s) apart within 45-day window
Policy POL9250571022	insurance_claims.policy_id	FD-07 FD-02 FD-07	1 day(s) apart within 45-day window
Account ACC6461174921	account_id (owner, beneficiary or payout account)	FD-03 FD-01	same moment
Application APP00235547	loan_applications.application_id	FD-05 FD-02	1.5 day(s) apart
Application APP07230798	loan_applications.application_id	FD-05 FD-02	1.5 day(s) apart
Application APP11540751	loan_applications.application_id	FD-05 FD-02	1.5 day(s) apart
Application APP84820769	loan_applications.application_id	FD-05 FD-02	1.5 day(s) apart
Customer CUS149029	customer_id	FD-05 FD-07 FD-02 FD-07 FD-04	5.4 day(s) apart
Customer CUS554152	customer_id	FD-05 FD-01 FD-07 FD-02 FD-07	1.5 day(s) apart
Customer CUS709143	customer_id	FD-05 FD-07 FD-02 FD-07 FD-04	5.4 day(s) apart
Customer CUS067889	customer_id	FD-07 FD-02 FD-07	1 day(s) apart

Consolidation explanation and split

5.4 Complete investigation workspace (requirement 4)

The case page brings together:

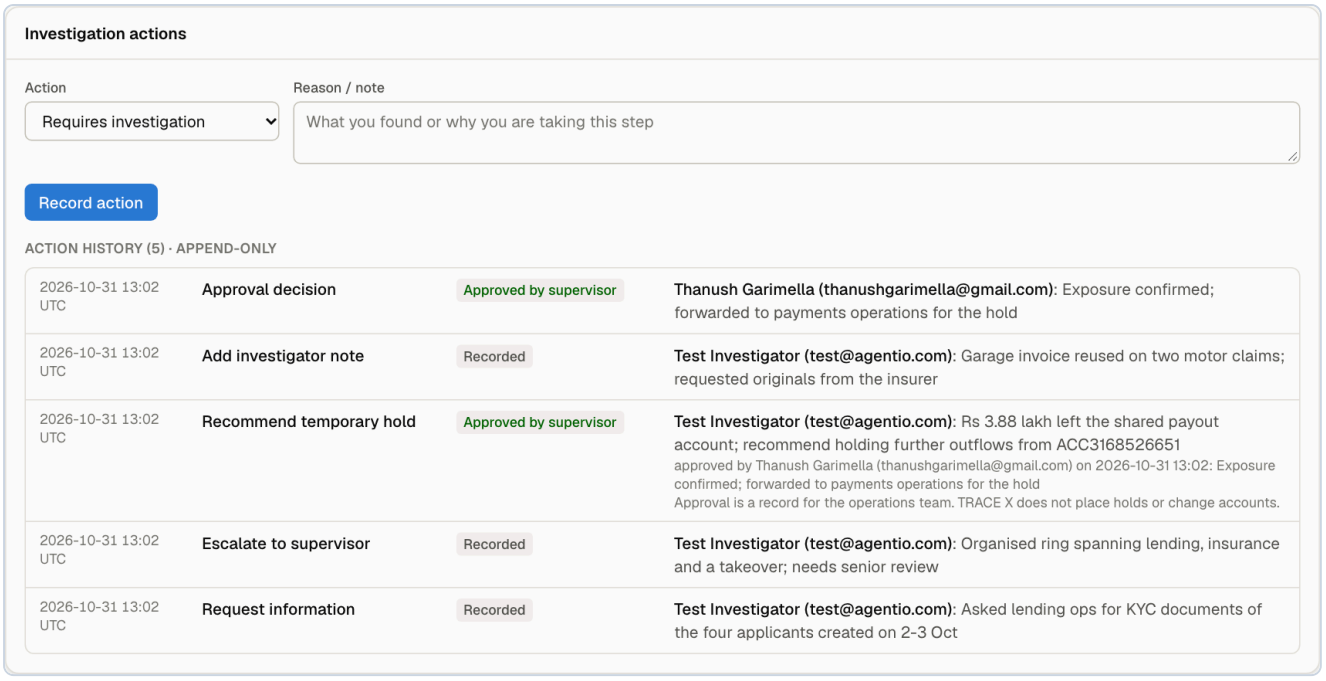
- the score strip and its breakdown;
- investigation actions;
- the agent investigation;
- the investigation brief;
- the graph-model view;
- the relationship graph;
- the consolidation explanation;
- the cross-system journey timeline;
- the full source records.



Case header and score strip

Actions: Requires Investigation, False Positive (reason required), Request Information, Escalate to Specialist, Escalate to Supervisor, Recommend Temporary Hold, Recommend Enhanced Verification, Close Case, Add Investigator Note and Record Investigation Conclusion (confirmed fraud, not fraud or inconclusive).

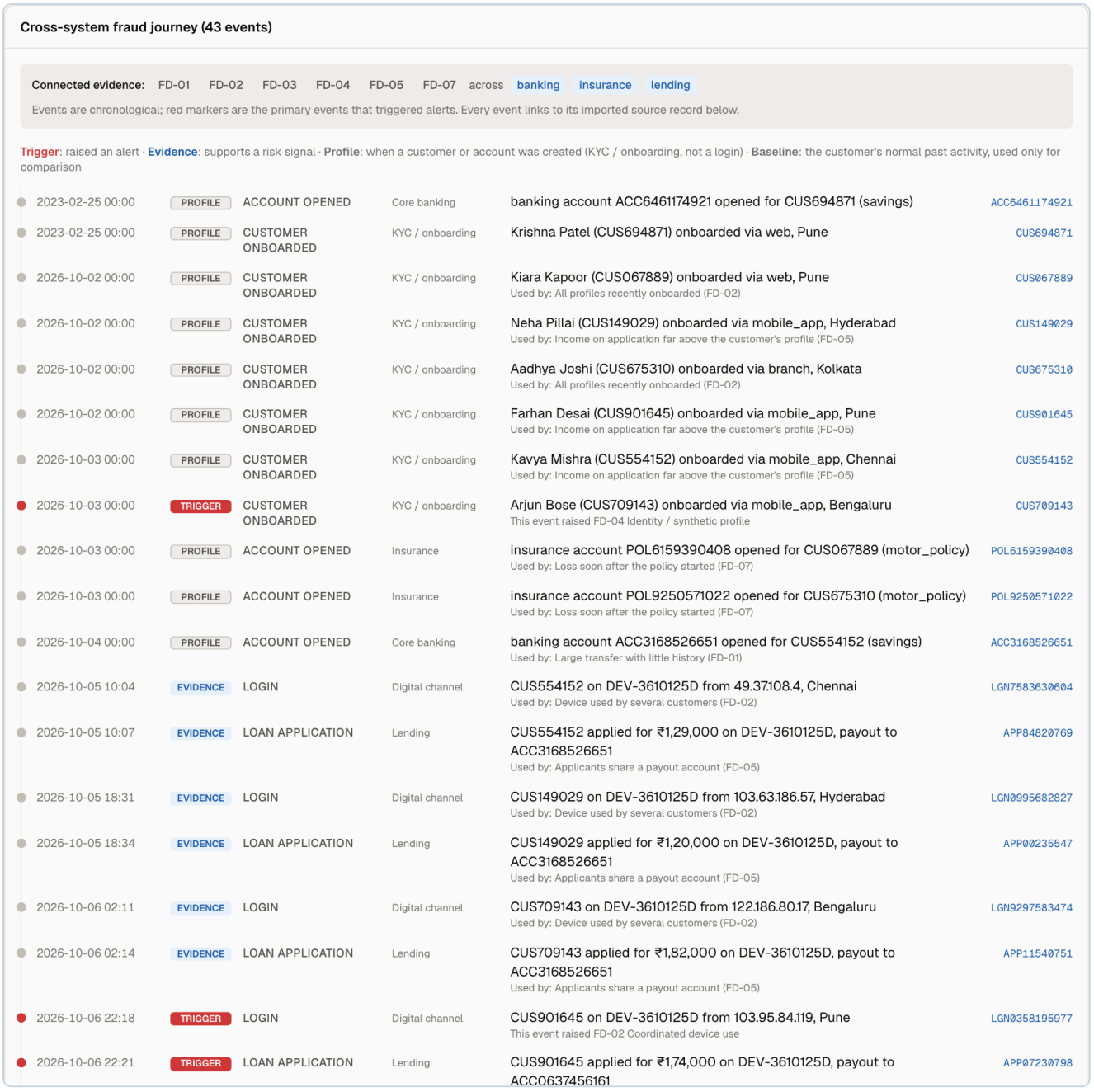
- **Recorded in full:** every action stores status, actor and timestamp in an append-only history.
- **Approval required:** hold and enhanced verification are customer-impacting, so they stay **recommendations** until a supervisor approves them. The approver must be an administrator other than the person who raised the request. Even an approval is only a record for operations; TRACE X never places holds or changes accounts.



Actions and approvals

Timeline: every event is labelled with its role and source system:

- **Trigger:** the event raised an alert.
- **Evidence:** it supports a signal.
- **Profile:** a customer or account was created (KYC, not a login).
- **Baseline:** the customer's normal history, used only for comparison and folded away.



Cross-system fraud journey (43 events)

Connected evidence: FD-01 FD-02 FD-03 FD-04 FD-05 FD-07 across banking insurance lending

Events are chronological; red markers are the primary events that triggered alerts. Every event links to its imported source record below.

Trigger: raised an alert · **Evidence**: supports a risk signal · **Profile**: when a customer or account was created (KYC / onboarding, not a login) · **Baseline**: the customer's normal past activity, used only for comparison

2023-02-25 00:00	PROFILE	ACCOUNT OPENED	Core banking	banking account ACC6461174921 opened for CUS694871 (savings)	ACC6461174921
2023-02-25 00:00	PROFILE	CUSTOMER ONBOARDED	KYC / onboarding	Krishna Patel (CUS694871) onboarded via web, Pune	CUS694871
2026-10-02 00:00	PROFILE	CUSTOMER ONBOARDED	KYC / onboarding	Kiara Kapoor (CUS067889) onboarded via web, Pune Used by: All profiles recently onboarded (FD-02)	CUS067889
2026-10-02 00:00	PROFILE	CUSTOMER ONBOARDED	KYC / onboarding	Neha Pillai (CUS149029) onboarded via mobile_app, Hyderabad Used by: Income on application far above the customer's profile (FD-05)	CUS149029
2026-10-02 00:00	PROFILE	CUSTOMER ONBOARDED	KYC / onboarding	Aadhya Joshi (CUS675310) onboarded via branch, Kolkata Used by: All profiles recently onboarded (FD-02)	CUS675310
2026-10-02 00:00	PROFILE	CUSTOMER ONBOARDED	KYC / onboarding	Farhan Desai (CUS901645) onboarded via mobile_app, Pune Used by: Income on application far above the customer's profile (FD-05)	CUS901645
2026-10-03 00:00	PROFILE	CUSTOMER ONBOARDED	KYC / onboarding	Kavya Mishra (CUS554152) onboarded via mobile_app, Chennai Used by: Income on application far above the customer's profile (FD-05)	CUS554152
2026-10-03 00:00	TRIGGER	CUSTOMER ONBOARDED	KYC / onboarding	Arjun Bose (CUS709143) onboarded via mobile_app, Bengaluru This event raised FD-04 Identity / synthetic profile	CUS709143
2026-10-03 00:00	PROFILE	ACCOUNT OPENED	Insurance	insurance account POL6159390408 opened for CUS067889 (motor_policy) Used by: Loss soon after the policy started (FD-07)	POL6159390408
2026-10-03 00:00	PROFILE	ACCOUNT OPENED	Insurance	insurance account POL9250571022 opened for CUS675310 (motor_policy) Used by: Loss soon after the policy started (FD-07)	POL9250571022
2026-10-04 00:00	PROFILE	ACCOUNT OPENED	Core banking	banking account ACC3168526651 opened for CUS554152 (savings) Used by: Large transfer with little history (FD-01)	ACC3168526651
2026-10-05 10:04	EVIDENCE	LOGIN	Digital channel	CUS554152 on DEV-3610125D from 49.37.108.4, Chennai Used by: Device used by several customers (FD-02)	LGN7583630604
2026-10-05 10:07	EVIDENCE	LOAN APPLICATION	Lending	CUS554152 applied for ₹1,29,000 on DEV-3610125D, payout to ACC3168526651 Used by: Applicants share a payout account (FD-05)	APP84820769
2026-10-05 18:31	EVIDENCE	LOGIN	Digital channel	CUS149029 on DEV-3610125D from 103.63.186.57, Hyderabad Used by: Device used by several customers (FD-02)	LGN0995682827
2026-10-05 18:34	EVIDENCE	LOAN APPLICATION	Lending	CUS149029 applied for ₹1,20,000 on DEV-3610125D, payout to ACC3168526651 Used by: Applicants share a payout account (FD-05)	APP00235547
2026-10-06 02:11	EVIDENCE	LOGIN	Digital channel	CUS709143 on DEV-3610125D from 122.186.80.17, Bengaluru Used by: Device used by several customers (FD-02)	LGN9297583474
2026-10-06 02:14	EVIDENCE	LOAN APPLICATION	Lending	CUS709143 applied for ₹1,82,000 on DEV-3610125D, payout to ACC3168526651 Used by: Applicants share a payout account (FD-05)	APP11540751
2026-10-06 22:18	TRIGGER	LOGIN	Digital channel	CUS901645 on DEV-3610125D from 103.95.84.119, Pune This event raised FD-02 Coordinated device use	LGN0358195977
2026-10-06 22:21	TRIGGER	LOAN APPLICATION	Lending	CUS901645 applied for ₹1,74,000 on DEV-3610125D, payout to ACC0637456161	APP07230798

Timeline

Investigation brief: a deterministic summary in which every sentence cites the records it uses, followed by next checks.

Investigation brief

WHAT HAPPENED

1. Cross-system organised fraud case, High risk (100/100), built from 8 alerts by 6 detector(s) across banking, insurance, lending.
   TXN7128359018 LGN0336197589 APP07230798 TXN0138030536 CLM50555786 LGN0358195977 CLM92436760 TXN7128359018 CUS709143
2. 7 customers involved, centred on CUS149029 (also CUS067889, CUS554152, CUS675310, CUS694871…). CUS067889 CUS149029 CUS554152 CUS675310 CUS694871 CUS709143
3. Strongest evidence: same supporting document on another claim (Document d4be2240c6eee7f4 is also attached to CLM50555786 (CUS067889, policy POL6159390408)); payout account used by other unrelated customers (Payout account ACC3168526651 is also used or owned by CUS067889, CUS554152, CUS149029, CUS709143); three or more profiles on one device (6 distinct profiles on one device).
   CLM92436760 CLM50555786 APP84820769 APP00235547 APP11540751 ACC3168526651 LGN0995682827 LGN7583630604
4. No legitimate context (household, established payee, normal range, known device) was found to offset the risk.
5. Activity spans 2023-02-25 to 2026-10-11; flagged events in order: customer onboarded on 2026-10-03 → login on 2026-10-06 → loan application on 2026-10-06 → motor claim on 2026-10-07.
   CUS709143 LGN0358195977 APP07230798 CLM50555786 TXN0138030536 CLM92436760

NEXT CHECKS

- ☐ Ask the provider for the original invoice and check it against both claims CLM50555786 CLM92436760
- ☐ Confirm with the bank who owns the shared payout account and freeze pending disbursements or settlements to it
  APP84820769 APP00235547 APP11540751
- ☐ Confirm the purpose of the unusually large transfer with the customer
  TXN7128359018 TXN7478132151 TXN1030086509 TXN2397745319
- ☐ Treat the transfer as possibly unauthorised until the customer confirms it
  LGN3881147899 LGN5069458486 LGN8719922367 LGN6547758230
- ☐ Request a recall of the transfer made right after the suspicious login TXN7128359018

Generated from this case's stored signals and records with a fixed template; no external model and no information beyond the evidence.

Investigation brief

Evidence: every source record, with its file and row of origin.

Supporting records (43)

CUSTOMERS (7)

Customer	Name	Phone	Email	Address	Born	National ID	Income	Onboarded	S
CUS694871	Krishna Patel	9332495942	krishna.patel24@outlook.com	181, Hill Road, Pune	1990-05-07	LAKPS8562C	₹12,96,000	2023-02-25	s
CUS554152	Kavya Mishra	7680141640	kavya.mishra436@rediffmail.com	356, Temple Street, Chennai	1980-11-06	GSFPA4568P	₹3,60,000	2026-10-03	s
CUS067889	Kiara Kapoor	8825588596	kiara.kapoor936@yahoo.co.in	229, Brigade Road, Pune	1995-05-11	TDGPK8958U	₹3,60,000	2026-10-02	s
CUS149029	Neha Pillai	9997915012	neha.pillai978@rediffmail.com	17, Gandhi Nagar Main Road, Hyderabad	1970-11-06	LDIPE9152V	₹3,60,000	2026-10-02	s
CUS675310	Aadhya Joshi	7350745001	aadhya.joshi784@gmail.com	358, Brigade Road, Kolkata	1978-02-15	GCKPP9373F	₹3,60,000	2026-10-02	s
CUS709143	Arjun Bose	+91 99979 15012	arjun.bose652@outlook.com	80, Brigade Road, Bengaluru	1960-08-18	VEKPU7383V	₹3,60,000	2026-10-03	s
CUS901645	Farhan Desai	7946873059	farhan.desai580@rediffmail.com	145, FC Road, Pune	1995-07-06	VRDPX6746A	₹3,60,000	2026-10-02	s

ACCOUNTS (4)

Account	Customer	Product	Type	Opened	Source
ACC6461174921	CUS694871	banking	savings	2023-02-25	seed-1/accounts.csv:497
ACC3168526651	CUS554152	banking	savings	2026-10-04	seed-1/accounts.csv:492
POL6159390408	CUS067889	insurance	motor_policy	2026-10-03	seed-1/accounts.csv:499
POL9250571022	CUS675310	insurance	motor_policy	2026-10-03	seed-1/accounts.csv:501

TRANSACTIONS (15)

Transaction	Time (UTC)	From	To	Amount	Channel	Type	Status	Source
TXN7128359018	2026-10-11 14:25	ACC6461174921	EXT3105786397	₹1,44,000	mobile	transfer	completed	seed-1/transactions.csv:4531
TXN7478132151	2026-08-25 16:48	ACC6461174921	ACC6427196368	₹15,940	mobile	transfer	completed	seed-1/transactions.csv:2408
TXN1030086509	2026-08-29 08:37	ACC6461174921	ACC6427196368	₹24,370	mobile	transfer	completed	seed-1/transactions.csv:2521
TXN2397745319	2026-09-02 12:15	ACC6461174921	MER-ECM-005	₹7,430	mobile	payment	completed	seed-1/transactions.csv:2979
TXN1417133216	2026-09-03 12:49	ACC6461174921	EXT7203030980	₹12,710	mobile	transfer	completed	seed-1/transactions.csv:3012
TXN5020116109	2026-09-05 18:25	ACC6461174921	ACC6427196368	₹11,030	mobile	transfer	completed	seed-1/transactions.csv:3067
TXN0624308929	2026-09-08 18:05	ACC6461174921	ACC6427196368	₹44,990	mobile	transfer	completed	seed-1/transactions.csv:3161
TXN7332863763	2026-09-10 16:51	ACC6461174921	EXT6199660746	₹23,490	mobile	transfer	completed	seed-1/transactions.csv:3229
TXN8662042485	2026-09-17 14:17	ACC6461174921	EXT1874471154	₹18,680	mobile	transfer	completed	seed-1/transactions.csv:3468
TXN3197753834	2026-09-27 14:27	ACC6461174921	EXT6199660746	₹34,440	upi	transfer	completed	seed-1/transactions.csv:3765

Evidence

5.5 Evidence management and structured case report (requirement 5)

The report can be downloaded as **PDF** or structured **JSON**. It contains:

- case details and the separate scores;
- the consolidation explanation and splits;
- every alert with its signals and mitigations;
- the timeline and graph relationships, each with its evidence;
- agent assessments, including tool calls and SQL;
- decisions, actions and approvals;
- an evidence index giving each record's source file, row and hash.

Every reference is re-resolved against the database when the report is built: the number of unresolved references is reported and tested to be zero. A SHA-256 integrity hash covers the report content.

TRACE X - Fraud Case Report

CASE-ca29ed0401 | ESCALATED | generated 2026-10-31T13:08:11.830Z by Test Investigator (test@agentio.com)

8 linked alerts across 6 detectors, 7 customers

Scores (kept separate)

Risk: High (100/100) Confidence: 100% Exposure: Rs 20,61,900 Priority: Critical (100)
Agent trust (Cross-system fraud): 0.034 from 2 validated decision(s); autonomy L1 - Assist
Weighted from 6 explicit components.

Why these alerts are one case

8 alerts from 6 detector(s) were grouped because they share 22 entities found in the source records.
- Account ACC3168526651 [account_id (owner, beneficiary or payout account)] links FD-05, FD-01, FD-07, FD-02 (5 alerts)
- Phone 9997915012 [customers.phone_norm] links FD-05, FD-07, FD-02, FD-04 (5 alerts) (within 45 days)
- Device DEV-3610125D [device_id] links FD-05, FD-07, FD-02 (4 alerts)
- Account EXT3105786397 [account_id (owner, beneficiary or payout account)] links FD-03, FD-01 (3 alerts)
- Claim CLM50555786 [insurance_claims.claim_id] links FD-07, FD-02 (3 alerts)
- Claim CLM92436760 [insurance_claims.claim_id] links FD-07, FD-02 (3 alerts)
- Policy POL6159390408 [insurance_claims.policy_id] links FD-07, FD-02 (3 alerts) (within 45 days)
- Invoice document d4be2240c6eee7f4 [insurance_claims.document_hash] links FD-07, FD-02 (3 alerts) (within 45 days)
- Policy POL9250571022 [insurance_claims.policy_id] links FD-07, FD-02 (3 alerts) (within 45 days)
- Account ACC6461174921 [account_id (owner, beneficiary or payout account)] links FD-03, FD-01 (2 alerts)
- Application APP00235547 [loan_applications.application_id] links FD-05, FD-02 (2 alerts)
- Application APP07230798 [loan_applications.application_id] links FD-05, FD-02 (2 alerts)

Alerts (9)

ALT-3b88ee0bc1 FD-06 High 100 8 linked alerts across 6 detectors, 7 customers
+ Strongest linked alert (+100): Account takeover (FD-03) scored 100: 5 failed logins then access on a new device from a new location for CUS694871
+ Independent detectors agree (+50): FD-01 Suspicious transfer, FD-02 Coordinated device use, FD-03 Account takeover, FD-04 Identity / synthetic profile, FD-05 Loan application fraud, FD-07 Insurance claim fraud
+ Spans several product systems (+10): banking, insurance, lending
ALT-2b4b5d574a FD-03 High 100 5 failed logins then access on a new device from a new location for CUS694871
+ Burst of failed logins (+21): 5 failed logins within 5 min
+ Successful login right after failures (+10): Successful login 3 min after the last failure
+ Login from a device never used before (+20): DEV-7BF6C325 was never used by this customer before (known: DEV-C4060AA1)
+ Login from a new IP address or location (+10): new IP 157.62.102.67, new location Chennai (usual: Pune)
+ Remote-access tool active (+12): Remote-access or screen-sharing tool was active in the session
ALT-3aa09007e3 FD-05 High 100 4 linked loan applications from 4 customers (Rs 6,05,000)
+ Applicants share a payout account (+30): ACC3168526651 receives payouts for 3 customers
+ Applications from the same device (+20): DEV-3610125D submitted applications for 4 customers
+ Same phone or email under different customers (+20): phone 9997915012 used by 2 customers
+ Many applications in a short window (+15): 4 linked applications within 7 days
+ Income on application far above the customer's profile (+15): APP84820769 states Rs 10,80,000 a year vs Rs 3,60,000 on the customer profile; APP00235547 states Rs 10,80,000 a year vs Rs 3,60,000 on the customer profile; APP11540751 states Rs 10,80,000 a year vs Rs 3,60,000 on the customer profile; APP07230798 states Rs 10,80,000 a year vs Rs 3,60,000 on the customer profile
ALT-09925fbf60 FD-01 High 97 Rs 3,87,900 from ACC3168526651 to EXT3105786397
+ High-value transfer (+25): Rs 3,87,900 is at or above the Rs 2,00,000 high-value threshold
+ Large transfer with little history (+15): Rs 3,87,900 from an account 3 days old with only 0 earlier transfers
+ First transfer to this beneficiary (+15): First transfer from ACC3168526651 to EXT3105786397 (Arjun Kaur Traders)
+ Very new account (+10): Account opened 3 days before this transfer
+ Account drained soon after inflow (+20): Rs 3,87,900 sent out within 24 h after Rs 4,31,000 came in during the previous 7 days (90%)
ALT-39a670cda2 FD-07 High 90 motor claim CLM50555786: Rs 4,60,000 by CUS067889
+ Same supporting document on another claim (+35): Document d4be2240c6eee7f4 is also attached to CLM92436760

TRACE X CASE-ca29ed0401 - page 1 of 5

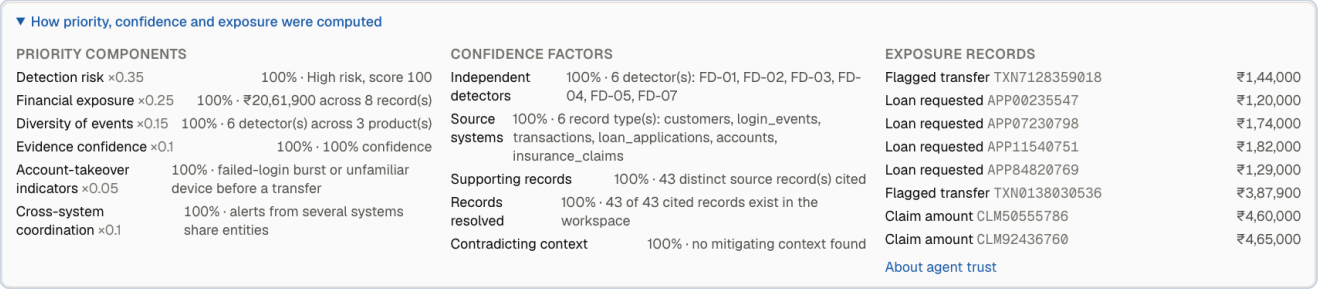
Case report PDF, page 1

5.6 Explainable prioritisation (requirement 6)

Five quantities are kept separate and never merged:

Quantity	Answers	Built from
Risk score	How suspicious is the evidence?	Detector signals – mitigations
Evidence confidence	How complete and corroborated is it?	Independent detectors, source systems, record count, records resolved, contradicting context
Financial exposure	How much money is at stake?	Flagged transfers, requested loans and claimed amounts, each record counted once
Investigation priority	What should be worked first?	Weighted: risk 35 %, exposure 25 %, diversity 15 %, confidence 10 %, takeover 5 %, cross-system 10 %
Agent trust	How reliable is the agent in this fraud category?	Analyst feedback (section 7)

A case with confidence below 45 % and no cross-system corroboration is marked **Needs More Evidence**, and its priority is capped instead of being escalated on risk alone.



Priority, confidence and exposure breakdown

6. Machine learning

6.1 Isolation Forest on transfers

A seeded Isolation Forest (100 trees, 256 samples) scores every outgoing transfer on 14 behavioural features, including:

- amount against the account's own history;
- hour against the customer's usual hour;
- whether the beneficiary is new and how many senders it has;
- channel rarity, account age and 24-hour velocity;
- outflow against recent inflow and amount against declared income.

The top 0.2 % adds a weighted signal to FD-01. The top 0.5 % that no rule caught becomes a Low FD-ML finding, which widens coverage without flooding the queue.

6.2 Graph neural network on the customer graph

Graph. Customers are joined when they share a device, phone, email, national ID, payout or beneficiary account, or invoice. Hubs are excluded. Edge weights are discounted by group size.

Features. 26 per-customer features computed from raw records. No rule-detector output and no labels are used. One feature, the highest Isolation Forest percentile among the customer's transfers, comes from the transfer anomaly model, so the GNN is independent of the rules but not of the Isolation Forest. They cover:

- tenure;
- devices and failed logins;
- session telemetry;
- transfer behaviour;
- payee-name mismatches;
- isolation-forest percentile;
- drain ratio;
- loans against income;
- claims against sum insured;
- device and account sharing.

Model. SIGN-style propagation concatenates the node's own features with its 1-hop and 2-hop neighbourhood features (Kipf-Welling normalisation). An MLP trained with Adam on weighted cross-entropy sits on top. Propagation is fixed rather than learned, a deliberate choice for small graphs on CPU-only serverless.

Evaluation protocol. Data is split by dataset, so no customer appears in two splits:

- **Train:** seeds 1–3 (999 customers, 51 fraud-linked).
- **Validation:** seed 101, used only to choose the model and threshold.
- **Sealed test:** seeds 201 and 202 (666 customers, 34 fraud-linked), scored once after selection.

Model	Validation AP	Sealed test AUC	Sealed test AP
Logistic regression, own features (no graph)	0.809	0.991	**0.886**
Logistic regression on graph features (SGC)	0.810	0.997	0.857
GNN: SIGN + MLP(16)	0.893	0.997	0.862
GNN: SIGN + MLP(32), selected	**0.980**	**0.997**	0.879

At the threshold chosen on validation (0.29), the GNN flags 36 of 666 sealed-test customers, with **precision 0.94 and recall 1.00**.

Reading these results honestly. The GNN clearly won on validation. On the sealed test it ties the no-graph baseline: the baseline's average precision is slightly higher, and the GNN's AUC is slightly higher. The graph's practical value is that it explains a customer's risk through their linked neighbours, which the investigator sees on the case page. Because fraud scenarios in the synthetic data are structurally obvious, every model scores high; these figures show the method generalises across datasets, not real-world accuracy.

Customers above the threshold with no actionable rule alert become **FD-GNN** monitored findings. Each is explained by its drivers (feature occlusion) and its graph neighbours.

Graph neural network: 7 of 7 customer(s) above threshold

Graph neural network: SIGN propagation + MLP(32), trained on seeds 1–3, threshold 0.29 chosen on validation seed 101. Sealed test (seeds 201–202): AUC 0.997, precision 0.944, recall 1. Model card

CUS067889	100% above threshold	No strong drivers Graph neighbours: CUS149029 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS709143 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS675310 (100%) via device DEV-3610125D, payout account ACC3168526651
CUS149029	100% above threshold	No strong drivers Graph neighbours: CUS709143 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS067889 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS675310 (100%) via device DEV-3610125D, payout account ACC3168526651
CUS675310	100% above threshold	No strong drivers Graph neighbours: CUS149029 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS709143 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS067889 (100%) via device DEV-3610125D, payout account ACC3168526651
CUS694871	100% above threshold	No strong drivers Graph neighbours: CUS554152 (100%) via beneficiary EXT3105786397
CUS709143	100% above threshold	No strong drivers Graph neighbours: CUS149029 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS067889 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS675310 (100%) via device DEV-3610125D, payout account ACC3168526651
CUS554152	100% above threshold	No strong drivers Graph neighbours: CUS149029 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS709143 (100%) via device DEV-3610125D, payout account ACC3168526651; CUS067889 (100%) via device DEV-3610125D, payout account ACC3168526651
CUS901645	100% above threshold	Driven by 2-hop: claim soon after policy start Graph neighbours: CUS149029 (100%) via device DEV-3610125D; CUS709143 (100%) via device DEV-3610125D; CUS067889 (100%) via device DEV-3610125D

Graph neural network on a case

Graph neural network

Selected by validation: **Graph neural network: SIGN propagation + MLP(32)**. It scores each customer from their own records and their neighbours in the customer graph (shared device, phone, email, national ID, payout or beneficiary account, invoice). Split by dataset so no customer appears in two splits; the test seeds were used once, after selection.

Train	Validation	Sealed Test
Seed 1, 2, 3	**Seed 101**	**Seed 201, 202**
999 customers, 51 fraud-linked	333 customers, 17 fraud-linked	666 customers, 34 fraud-linked

Candidates (average precision is the fair metric with 5% positives)

Model	Train AP	Validation AUC	Validation AP	Test AUC	Test AP
Logistic regression, own features (no graph)	0.999	0.973	0.809	0.991	0.886
Logistic regression on graph features (SGC)	1	0.994	0.81	0.997	0.857
Graph neural network: SIGN propagation + MLP(16)	1	0.996	0.893	0.997	0.862
Graph neural network: SIGN propagation + MLP(32) ✓	1	0.999	0.98	0.997	0.879

Operating threshold 0.29, set on validation for precision ≥ 0.8. On the sealed test seeds it flags 36 customers: precision 0.944, recall 1.

Without the enriched fields (recent SIM swap, remote-access tool seen, emulator / rooted device seen, VPN / proxy share, max behaviour anomaly, payee-name mismatch share) the same architecture reaches test AP 0.862 vs 0.879.

Synthetic data with planted scenarios: these numbers show the method generalises across datasets, not real-world accuracy. Train AP of 1.0 shows the model can memorise; held-out numbers are the ones that count.

Model card on the Evaluation page

7. Adaptive agentic investigator

7.1 The agent team

Step	Agent	What it does
1	Orchestrator	Reads the case and writes a brief, the entities to focus on, open questions and a hypothesis budget
2	Hypothesis agent	Proposes competing explanations, fraud and legitimate, citing case IDs; applies lessons from past corrections
3	SQL agent	Writes read-only SQL to test each hypothesis; failed queries get one self-repair round with the error message
4	Reviewer agent	Judges each hypothesis from the query results; returns verdicts, confidence, cited IDs, an investigation-quality score, recommended actions and missing information
5	Code verification	Checks every cited ID against query results and the database; drops anything not found; downgrades a "supported" verdict that has no verified citation
6	Feedback-learning agent	Compares the conclusion with the analyst's verdict and writes a short, general lesson that is retrieved into future prompts for the same detector signature

Safety boundaries:

- **Narrow SQL access:** the SQL agent sees only source tables and graph scores, never case decisions, feedback or lessons, so it cannot read the answer it is graded on. Each query must be a single SELECT; it runs inside an outer SELECT with a 50-row cap, and write keywords are blocked.
- **Approval stays with humans:** customer-impacting recommendations are always marked as requiring human approval.
- **Fallback:** without an API key, or on any model error, a deterministic tool-using agent runs instead, and the panel says so.

Adaptive investigation agent

Re-run multi-agent investigation | Run deterministic agent

Orchestrator → hypothesis → SQL → reviewer agents (OpenAI), then code verifies every cited ID against the database. Falls back to the deterministic tool agent without a key. Agents read data only and write their own assessment.

Earned trust for **Cross-system fraud**: **0.034** from 2 validated decision(s) · L1 — Assist. for L2: 6 more decision(s); trust 0.03 → 0.45. Trust is separate from this case's risk, confidence and exposure.

Suspected category	Priority	Claims verified against records
Cross-system fraud	**Critical**	**12/12**
Evidence confidence 70%	Weighted from 6 explicit components.	none dropped · run 2026-10-31 13:02 by Test Investigator (test@agentio.com)

AGENT WORKFLOW · GPT-4O-MINI-2024-07-18 · REVIEWER SCORE 80% · 15579 TOKENS

▸ Orchestrator 2.2 s → ▸ Hypothesis 3.6 s → ▸ Sql 3.3 s → ▸ Sql 2.1 s → ▸ Reviewer 5.2 s → code verification: 12/12 citations found

Applied 5 lesson(s) learned from earlier analyst corrections: *When assessing suspicious transfers, ensure to evaluate the context and historical patterns of accounts involved, as automated systems may misinterpret isolated transactions without considering broader account behavior.* · *When assessing suspicious transfers, prioritize patterns of large, unusual transactions, especially when linked to new or unverified accounts, as they often indicate potential fraud.*

▸ SQL written by the SQL agent (2 queries, read-only, source tables only)

HYPOTHESES · REVIEWER CONFIDENCE

Loan fraud 95% · 1 supporting, 0 contradicting

\+ The linked loan applications from multiple customers indicate a coordinated effort to defraud the lending system, utilizing shared payout accounts and devices.. Reviewer: The investigation revealed multiple loan applications from different customers using the same device and payout account, indicating a coordinated effort to defraud the lending system. [CUS149029, CUS554152, CUS709143, ACC3168526651, DEV-3610125D] CUS149029 CUS554152 CUS709143 ACC3168526651

Account takeover 90% · 1 supporting, 0 contradicting

Likely legitimate 2% · 0 supporting, 1 contradicting

RECOMMENDED NEXT ACTIONS (A HUMAN DECIDES)

Requires investigation internal
The findings indicate potential fraud across multiple systems, necessitating a deeper investigation into the linked accounts and activities.

MISSING INFORMATION

• Documentation supporting legitimacy of claims and transfers: To fully assess the legitimacy of the high-value transfers and claims, documentation is needed to confirm the business purposes of the transactions.

TOOL CALLS (7)

1. `orchestrator_agent` · Read the case and plan the investigation
gpt-4o-mini-2024-07-18, 4859 tokens, 2210 ms

2. `hypothesis_agent` · Generate fraud hypotheses from the case records
gpt-4o-mini-2024-07-18, 5161 tokens, 3588 ms

3. `sql_agent` · Write SQL to test each hypothesis
gpt-4o-mini-2024-07-18, 1021 tokens, 3278 ms

4. `sql_agent` · Write SQL to test each hypothesis
gpt-4o-mini-2024-07-18, 804 tokens, 2124 ms

5. `run_sql` · Identify unauthorized access and suspicious transfers for CUS694871.
0 row(s)

6. `run_sql` · Detect coordinated loan applications using shared payout accounts and devices.
3 row(s) after self-repair

7. `reviewer_agent` · Review the evidence and score the investigation
gpt-4o-mini-2024-07-18, 3734 tokens, 5245 ms

Your review of this recommendation

(•) Accept | ( ) Correct the category | ( ) Reject (not fraud / wrong)

Submit review

Agent investigation panel

▾ SQL written by the SQL agent (2 queries, read-only, source tables only)

0 rows H1: Identify unauthorized access and suspicious transfers for CUS694871.

```
SELECT t.transaction_id, t.account_id, t.amount, t.timestamp, le.device_id, le.ip_address, le.geo FROM transactions t JOIN login_events le ON t.account_id = le.customer_id WHERE le.customer_id = 'CUS694871' AND le.outcome = 'failure' AND le.timestamp > (SELECT MAX(timestamp) FROM login_events WHERE customer_id = 'CUS694871') AND le.device_id NOT IN (SELECT device_id FROM login_events WHERE customer_id = 'CUS694871') LIMIT 50
```

3 rows self-repaired H2: Detect coordinated loan applications using shared payout accounts and devices.

```
SELECT la.application_id, la.customer_id, la.device_id, la.payout_account FROM loan_applications la WHERE la.payout_account IN (SELECT payout_account FROM loan_applications WHERE customer_id IN ('CUS067889', 'CUS149029', 'CUS554152', 'CUS675310', 'CUS694871', 'CUS709143', 'CUS901645') GROUP BY payout_account HAVING COUNT(DISTINCT customer_id) > 1) AND la.device_id IN (SELECT device_id FROM loan_applications WHERE customer_id IN ('CUS067889', 'CUS149029', 'CUS554152', 'CUS675310', 'CUS694871', 'CUS709143', 'CUS901645') GROUP BY device_id HAVING COUNT(DISTINCT customer_id) > 1) LIMIT 50
```

SQL written by the SQL agent

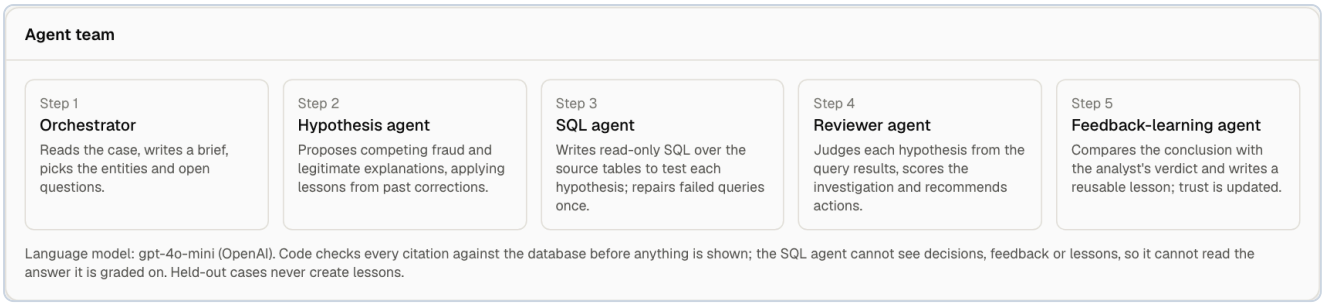
▼ Reviewer 5.2 s


```
{
  "hypotheses": [
    {
      "index": 0,
      "verdict": "supported",
      "confidence": 0.9,
      "cited_ids": [
        "CUS694871",
        "DEV-7BF6C325",
        "TXN7128359018"
      ],
      "rationale": "The investigation confirmed unauthorized access to CUS694871's account with
multiple failed logins followed by a successful login from a new device and location, supporting
the hypothesis of account takeover."
    },
    {
      "index": 1,
      "verdict": "supported"
```


Reviewer agent output

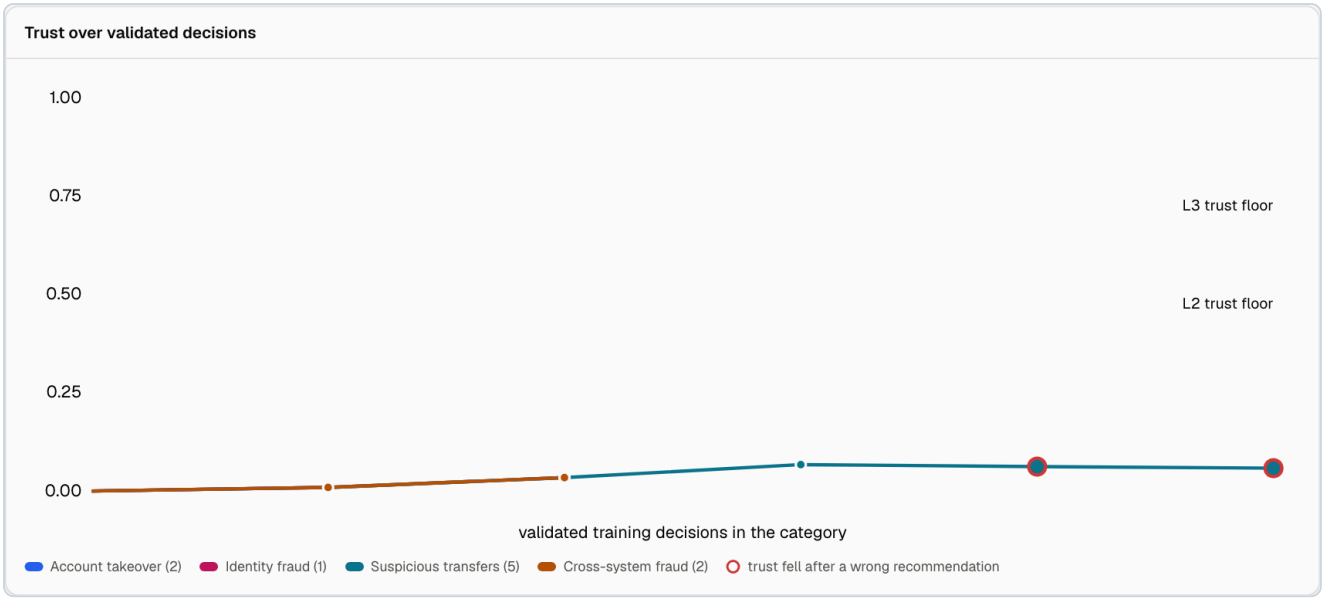
7.2 Agent Trust & Autonomy Engine

- **Per-category trust.** Trust is tracked separately for account takeover, shared-device coordination, identity fraud, loan fraud, insurance fraud, suspicious transfers and cross-system fraud.
- **Starts at zero.** Every category begins with zero trust and zero autonomous permissions.
- **Feedback capture.** Every analyst review records whether the recommendation was accepted, corrected (with the right category) or rejected, plus the reasoning and evidence IDs. Evidence IDs are validated against the database.
- **Model.** For each category, Beta(1 + s, 1 + f), where accepted recommendations add their evidence-grounding ratio to s, and corrected or rejected ones add 1 to f.
- **Trust score.** The lower 5 % credible bound, measured above the uninformed prior, multiplied by a volume discount until 20 decisions exist. Correct decisions raise it, incorrect ones lower it, and a small sample keeps it low.
- **Held out.** 25 % of cases, chosen by a hash of the case ID, are held out: they validate trust but never change it or create lessons.
- **Autonomy levels.**
  - **L0 (Observe):** no validated decisions; the agent analyses and a human approves everything.
  - **L1 (Assist):** at least one validated decision; evidence-backed suggestions.
  - **L2 (Recommend):** at least 8 decisions and trust of 0.45 or more; prioritised actions that a human confirms.
  - **L3 (Limited autonomy):** at least 20 decisions, trust of 0.7 or more, and at least 5 held-out decisions at 80 % or better accuracy. Only allow-listed, reversible, internal actions: label case, route to queue, request analyst review.
- **Controls.** Administrators can only tighten the thresholds, within safe floors. A global kill switch stops all autonomous actions.
- **Never autonomous.** Holds, blocks, account restrictions, fund transfers and final adverse decisions.



Agent team

The trust curve below is computed from real feedback. Red rings mark points where trust fell after a wrong recommendation.



Trust over validated decisions

Lessons learned from analyst feedback (10)

2026-10-31 13:02 UTC · FD-01 · agent said Suspicious transfers, analyst said Not fraud · written by the feedback-learning agent
Prioritize direct customer confirmation and name verification over account age and transaction history when assessing transfer legitimacy.

2026-10-31 13:02 UTC · FD-01+FD-02+FD-03+FD-04+FD-05+FD-07 · agent said Cross-system fraud, analyst agreed · written by the feedback-learning agent
In cases of cross-system fraud, prioritize patterns of shared payout accounts and device usage over isolated incidents of account takeover, as they indicate coordinated fraudulent activity.

2026-10-31 13:01 UTC · FD-01 · agent said Suspicious transfers, analyst said Not fraud · written by the feedback-learning agent
When assessing suspicious transfers, ensure to evaluate the context and historical patterns of accounts involved, as automated systems may misinterpret isolated transactions without considering broader account behavior.

2026-10-31 13:01 UTC · FD-01 · agent said Suspicious transfers, analyst agreed · written by the feedback-learning agent
When assessing suspicious transfers, prioritize patterns of large, unusual transactions, especially when linked to new or unverified accounts, as they often indicate potential fraud.

2026-10-31 13:01 UTC · FD-01+FD-08 · agent said Suspicious transfers, analyst agreed · written by the feedback-learning agent
Prioritize reactivation timelines in suspicious transfer cases; prolonged inactivity followed by sudden activity can indicate potential fraud.

2026-10-31 13:01 UTC · FD-01 · agent said Suspicious transfers, analyst agreed · written by the feedback-learning agent
When detecting suspicious transfers, prioritize high-risk flags from detectors like FD-01, especially when significant amounts are involved, as they indicate potential fraud.

2026-10-31 13:01 UTC · FD-04 · agent said Identity fraud, analyst agreed · written by the feedback-learning agent
When multiple profiles share critical identifiers like national ID and email, prioritize this as a strong indicator of identity fraud, as it consistently correlates with fraudulent activity.

2026-10-31 13:01 UTC · FD-01+FD-03 · agent said Account takeover, analyst agreed · written by the feedback-learning agent
Prioritize failed login attempts followed by new device access as strong indicators of account takeover, while also considering suspicious transfer patterns to ensure comprehensive risk assessment.

2026-10-31 13:01 UTC · FD-01+FD-02+FD-03+FD-04+FD-05+FD-07 · agent said Cross-system fraud, analyst agreed · written by the feedback-learning agent
Prioritize alerts with the highest risk scores, like suspicious transfers and account takeovers, especially when multiple indicators align, to strengthen fraud detection accuracy.

2026-10-31 13:01 UTC · FD-01+FD-03 · agent said Account takeover, analyst agreed · written by the feedback-learning agent
When assessing account takeovers, prioritize high-risk indicators like new device access from unfamiliar locations, especially when combined with suspicious transfer patterns.

Lessons learned from analyst feedback

Autonomy levels and permitted actions

Level	Condition	Allowed behaviour
L0 — Observe	No validated trust	Analyzes cases. A human approves every decision.
L1 — Assist	At least one validated decision	Generates evidence-backed suggestions.
L2 — Recommend	Enough decisions and trust above the L2 floor	Proposes prioritised actions; a human confirms each.
L3 — Limited autonomy	Volume, trust and held-out accuracy all above floor	Only labels cases, routes them to a queue and requests analyst review. All reversible and logged.

Current thresholds: L1 ≥ 1 decision; L2 ≥ 8 decisions and trust ≥ 0.45; L3 ≥ 20 decisions, trust ≥ 0.7, ≥ 5 held-out decisions at ≥ 80% accepted.
Autonomous allowlist: `LABEL_CASE`, `ROUTE_TO_QUEUE`, `REQUEST_ANALYST_REVIEW`. Never autonomous: holds, transaction blocks, account restrictions, fund transfers, final adverse decisions.
Kill switch: **off**

Autonomy levels and permitted actions

Administrator controls

Engage kill switch — Autonomy is only possible in categories that have earned L3.

L3 min training decisions (≥5): 20
L3 min held-out decisions (≥3): 5
Save thresholds — Trust floors stay L2 0.45, L3 0.7; held-out accuracy ≥ 80%.

Simulated analyst: answer-key seeds loaded in the workspace: 3
Replay answer key as feedback

For evaluation and demos only. Each case is judged against the bundled answer key; real analyst decisions stay separate (toggle below).

Administrator controls: kill switch, thresholds, answer-key replay

The feedback loop in practice. In the staged demo an analyst accepted the agent's "Cross-system fraud" conclusion, and trust rose from 0.009 to 0.034. A supervisor rejected a transfer recommendation because the customer confirmed it and the payee name matched; trust fell from 0.062 to 0.058, and the feedback-learning agent wrote: "*Prioritize direct customer confirmation and name verification over account age and transaction history when assessing transfer legitimacy.*"

7.3 Agent evaluation

Sealed benchmark. 19 cases from seeds 201 and 202 were scored against their answer keys. These cases were never used for training, tuning or lessons.

Agent	Correct	Accuracy	Citation grounding
Deterministic tool agent	18 / 19	94.7 %	1.00
LLM agent team (gpt-4o-mini)	18 / 19	94.7 %	0.99

Both agents miss the same case, one the answer key marks as not fraud. On this data the LLM team **matches but does not beat** the deterministic agent on category accuracy. What it adds is investigative depth: competing hypotheses, the SQL it ran, a reviewer score and lessons. The Evaluation page also reports, from live feedback: agent correctness, analyst override rate, trust calibration by confidence band, false-positive handling, grouping quality (split rate) and autonomous-action safety (allow-list violations, customer-impacting actions taken by the agent).

Agent evaluation

Computed from 12 validated decision(s): 2 by analysts, 10 by the labelled answer-key replay. With few decisions these rates are indicative only.

Sealed-seed benchmark (seeds 201, 202, never used for training, tuning or lessons)

Agent	Cases	Correct	Accuracy	Citation grounding	Fallbacks
Deterministic tool agent	19	18	94.7%	1	0
LLM multi-agent (gpt-4o-mini)	19	18	94.7%	0.99	0

Generated 2026-10-31 by scripts/eval-agent.ts. Agreement with the answer key on case category; the miss on both is a case the key lists as not fraud.

AGENT CORRECTNESS
83.3%
10 accepted · 0 corrected · 2 rejected

ANALYST OVERRIDE RATE
50%
all feedback: 16.7%

HELD-OUT ACCURACY
100%
2 held-out decision(s), never used to learn

AUTONOMOUS-ACTION SAFETY
0 violations
0 autonomous action(s); customer-impacting by agent: 0

Trust calibration: does stated confidence match accuracy?

Agent confidence	Decisions	Accepted
0–50%	1	100%
50–80%	5	80%
80–100%	6	83.3%

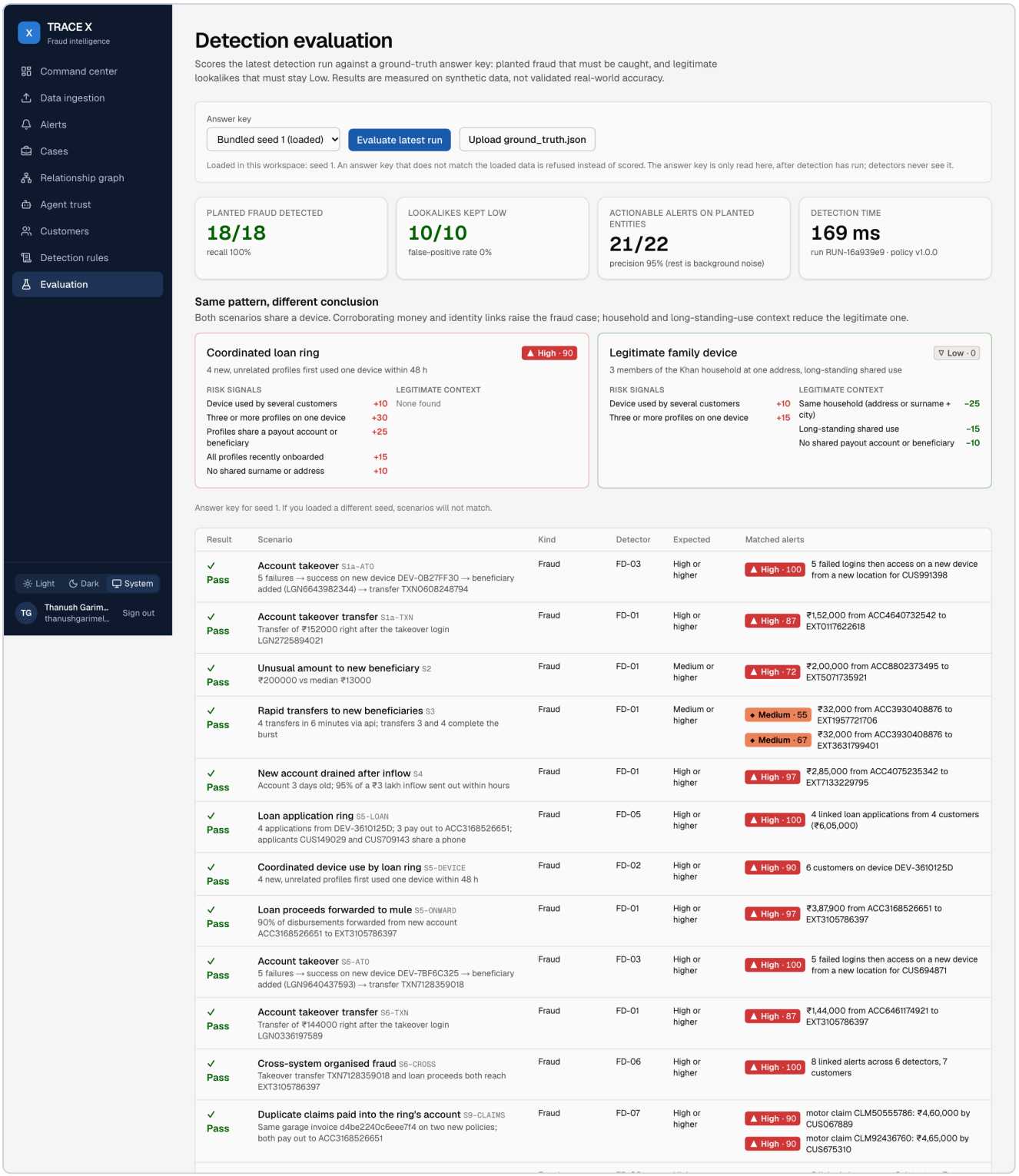
False-positive handling and grouping quality

Cases judged not fraud	2
…of which the agent had flagged as likely legitimate	0 (0%)
Cases / analyst splits	10 / 0 (0% split rate)

Per-category trust and held-out validation

Category	Training decisions	Trust	Level	Held-out n	Held-out accepted
Account takeover	2	0.034	L1 — Assist	0	—
Shared-device coordination	0	0.000	L0 — Observe	0	—
Identity fraud	1	0.009	L1 — Assist	0	—
Loan fraud	0	0.000	L0 — Observe	0	—
Insurance fraud	0	0.000	L0 — Observe	1	100%
Suspicious transfers	5	0.058	L1 — Assist	1	100%
Cross-system fraud	2	0.034	L1 — Assist	0	—

Agent evaluation



Detection evaluation against an answer key

8. What we built beyond the brief

1. **Enriched data fields that real banks have.**
    - **Login events:** SIM-swap age, remote-access or screen-share tool, emulator or rooted device, VPN or proxy, session length, and a behavioural-biometric anomaly score.
    - **Transfers:** the verify-payee (beneficiary name check) result, which RBI's 2025 rules made available for RTGS and NEFT and NPCI's lookup provides for UPI and IMPS.

    They are optional columns with header aliases and validation, used by the takeover and transfer rules and by the graph model. When they are missing, everything still works. In BFSI these are the strongest signals for account takeover and authorised-push-payment fraud, and they rarely appear in fraud demos.

2. **Three separate opinions, never averaged into one.** Explainable rules, an Isolation Forest and a graph neural network each give their own verdict, so disagreement stays visible. The GNN uses no rule output; it does take the Isolation Forest percentile as one of its 26 inputs.

3. **Agents that cannot invent evidence.** The LLM proposes, and code checks every citation. The SQL agent cannot see the human decisions it is graded on.
4. **Learning with a proper held-out discipline.** The GNN is trained on seeds 1–3, tuned on 101 and tested once on sealed 201 and 202. Agent trust is validated on held-out cases, and held-out cases never teach the agent.
5. **Autonomy that is earned per category, can be revoked, and is bounded by allow-list and kill switch.** Customer-impacting actions are never autonomous at any level.
6. **Analyst corrections survive re-detection.** Splits persist, superseded cases keep their decisions, and every record is append-only, enforced by database triggers.
7. **Every number is computed.** Trust curves, evaluation metrics, exposure and priority come from stored data. Nothing on screen is typed in.
8. **Transparency for a shared workspace.** The activity log shows who loaded, reset, decided, approved or changed autonomy settings, and the dashboard shows which dataset and detection run are on screen.

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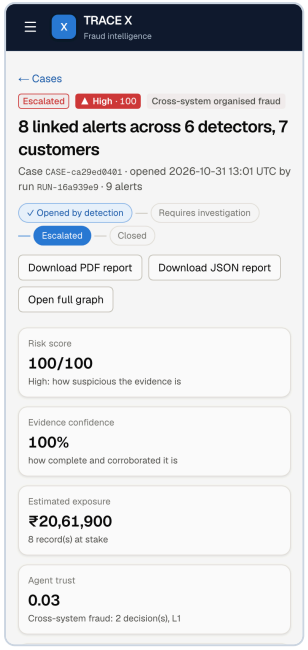
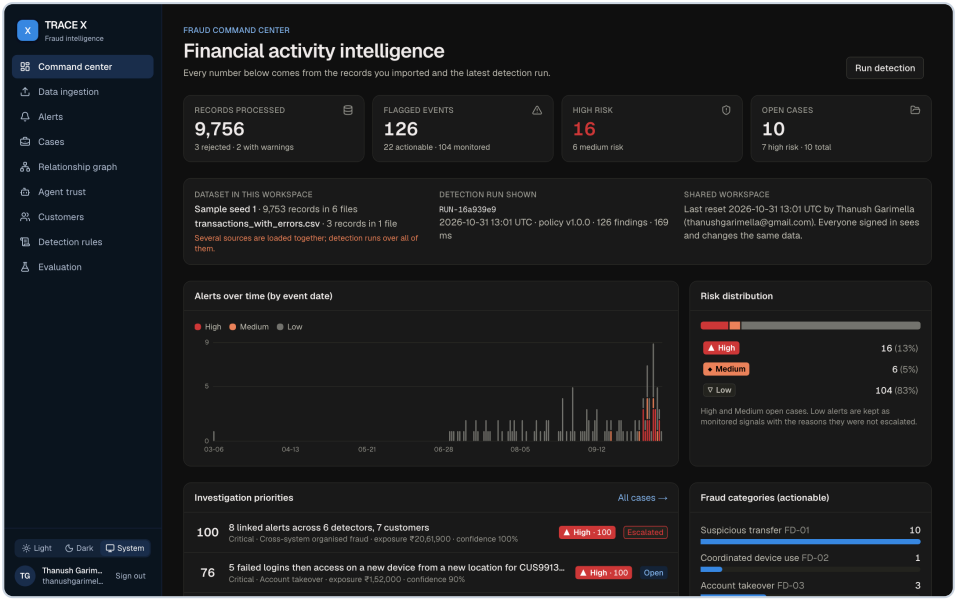
9. Security and governance

- **Access control:** HMAC-signed session cookies, scrypt-hashed passwords, and role checks on every mutating route. The actor is always taken from the session, never from the request body.
- **Approvals:** customer-impacting approvals require an administrator other than the requester. Approval decisions are single-use and append-only.
- **Secrets:** stored only in Vercel environment variables (`TURSO_DATABASE_URL`, `TURSO_AUTH_TOKEN`, `TRACEX_SESSION_SECRET`, `OPENAI_API_KEY`). The app refuses to start on Vercel without a durable database and a strong session secret.

- **Guarded agent SQL:** single statement, SELECT only, an allow-listed set of tables, an outer row cap and no access to decisions or feedback.
- **Uploads:** capped at 4.5 MB; the end-to-end scripts refuse to reset a hosted workspace without an explicit flag.

10. Responsive and accessible interface

The interface works at phone width, with a 16 px gutter and no horizontal scroll. Every page supports light and dark themes. Filters and inputs have labels, the graph has a text description for screen readers, and status is never shown by colour alone.

Mobile case view	Dark theme
	

11. Testing and verification

Suite	Count	Result
Unit and integration tests (`npm test`, in-memory databases)	111 tests in 10 files	**All pass**
Lint (`npx eslint`) and type check (`npm run typecheck`)	—	**Clean**
Production build (`npx next build`)	—	**Succeeds**
Level 1 end-to-end (`node scripts/smoke.mjs <url> 1`)	20 checks	**All pass**
Level 2 end-to-end (`node scripts/smoke-l2.mjs <url> 1`), with the real OpenAI key	32 checks	**All pass**
Detection evaluation, seeds 1, 2, 3, 101, 201 and 202	18/18 fraud, 10/10 lookalikes each	**All pass**
GNN training and sealed test (`npx tsx scripts/train-model.ts`)	see section 6.2	Reproducible
Agent sealed benchmark (`npx tsx scripts/eval-agent.ts`)	19 cases	18 / 19

What the Level 2 tests prove:

1. Correlation works through every required entity type, honours the time window and attenuates hubs.
2. Every graph edge resolves to source records.
3. Grouping is explained, analyst splits persist across re-runs, and an invalid split is refused.
4. All actions are recorded; history is immutable (UPDATE and DELETE are rejected by triggers); a reason is required.
5. Customer-impacting recommendations need a different, authorised approver and are never executed.
6. Exposure counts each record once; low-evidence cases become Needs More Evidence.
7. The PDF and JSON reports have zero unresolved references.
8. Agent output is structured, and every displayed claim resolves to records; a fabricated claim is dropped.
9. Correct feedback raises trust and incorrect feedback lowers it, in memory and in the database.
10. Categories without feedback stay at L0; L3 needs volume, trust and held-out accuracy together.
11. Crossing the L3 threshold allows only allow-listed, reversible actions; the kill switch blocks them.
12. The SQL guard rejects writes and hidden tables.
13. The multi-agent pipeline drops fabricated IDs, repairs failed SQL and filters disallowed actions.
14. Lessons come only from training-split feedback.
15. The GNN ranks fraud-linked customers well on a sealed seed.
16. All 64 Level 1 tests still pass.

12. Five-minute demonstration

1. **Command center (30 s).** Show the dataset and detection run on screen and the investigation priorities. Open the top Critical case, "8 linked alerts across 6 detectors, 7 customers".

2. **Scores (30 s).** Walk through the score strip: risk, confidence, exposure (₹20.6 lakh), agent trust and priority, kept separate. Open the breakdown.
3. **Graph (60 s).**
    - Click the shared payout account to see the loan disbursements and claim payouts behind it.
    - Click the external mule account, which links the loan ring to an unrelated customer's takeover.
    - Filter to suspicious links, then jump from an edge to its timeline event.
4. **Consolidation and correction (45 s).** Show the shared entities and matching fields. Split the takeover alert into its own case with a reason, re-run detection, and show that it stays split.
5. **Agent team (60 s).**
    - Run the multi-agent investigation and walk through orchestrator, hypothesis, SQL and reviewer.
    - Show the SQL it ran and the 12/12 verified citations.
    - Reject or correct the recommendation, then show the trust change and the lesson it wrote.
6. **Human authority (30 s).** As the analyst, recommend a temporary hold. As the administrator, approve it, and show that it is recorded but nothing is executed.
7. **Autonomy (45 s).** On Agent Trust, run the labelled answer-key replay, point to a red ring where trust fell, lower the L3 thresholds to their floor, and show the allow-listed autonomous actions in the audit log. Then engage the kill switch.

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13. Limitations and next steps

Limitations:

- **Synthetic data:** all datasets are synthetic with planted scenarios. Held-out results show that the methods generalise across datasets, not that they would reach this accuracy on real data. Validation on a bank's labelled history is the next step.
- **Fixed graph propagation:** the GNN does not learn its neighbourhood aggregation. GraphSAGE or GAT become worthwhile at bank scale with GPU training.
- **LLM variability:** LLM output varies from run to run. Verification bounds what reaches the screen but cannot make the reasoning deterministic.
- **Trust needs volume:** reaching L2 or L3 needs many human decisions per category. The answer-key replay is a labelled simulated analyst for demonstrations and is shown separately from human feedback.
- **One shared workspace:** a production deployment would add tenant isolation, SSO and an operations-system integration for approved recommendations.

Next steps:

- **Model hardening:** calibrate the GNN with temporal splits, and add drift monitoring on model inputs.
- **Data and integration:** connect live telemetry from device-intelligence SDKs and payee checks from the payment switch, and add case assignment and SLAs.